

Highlights

A Cartesian multipole hierarchy for the voxel: closed-form single-site T -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes

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- A Cartesian multipole hierarchy gives the cube's single-site T -matrix in closed form
- A voxel needs two bases: one for the medium and one for the wavefield
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T -matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

A Cartesian multipole hierarchy for the voxel: closed-form single-site T -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes

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Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$, with leading error $c_p(kd)^{2p+2}$ in closed form; the orders hold for SV and SH waves and at oblique incidence. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$: the two bases must be matched, and the error is known from the medium before anything is solved. In a plane-stratified background the orders persist: the background changes the field the voxels must represent, not how well they represent it. The single site is the first member of a

Cartesian multipole hierarchy, an elastodynamic extension of the equivalent inclusion method, whose coefficients are moments of the Green's tensor over the cube, in closed form on four constants. Its blocks gain accuracy in pairs: with third gradients it is 300 times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order on a graded sphere.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background,

16 which the model prescribes. The other is the *wavefield*: the displacement and
 17 strain inside the voxel, which the calculation must find. Each is expanded
 18 in a basis on the cell, and the two bases are separate choices. Here both
 19 are Legendre polynomials, the wavefield to degree p and the contrast to
 20 degree r ; the usual scheme, a uniform field in a cell of constant properties,
 21 is $p = r = 0$. Neither choice is new in itself. In higher-order volume integral
 22 methods for electromagnetics the material within an element is represented
 23 by polynomials whose orders are independent of those of the field [1], as it
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What
 25 has been missing is the law that relates the two bases: the order a given pair
 26 attains, and the error it leaves. The answer decides the cost of everything
 27 built on it, because a voxel that represents its contents more accurately can
 28 be larger, and the number of voxels, not the size of each one's state, sets the
 29 cost of the multiple-scattering solve.

30 *Why the question is hard to answer.* For the media such a scheme is built for,
 31 there is no exact solution. Its error cannot be measured against the answer.
 32 Comparison with another numerical method mixes the errors of both. And a
 33 refinement study can mislead, because a voxel scheme can converge smoothly
 34 to the wrong limit. §4 shows how that happens with the most natural choice
 35 of coupling, the Green's tensor evaluated between voxel centres: it carries
 36 a local term whose size per unit volume does not shrink as the voxels do,
 37 and unless the single voxel's response is matched to cancel it, refinement
 38 converges to the wrong answer without revealing it.

39 *The method: a layer of voxels.* This paper therefore answers the question
 40 in the simplest configuration in which a voxel scheme runs in full (every

41 voxel, every coupling, the whole multiple-scattering series, at normal and
 42 at oblique incidence) and the exact answer is nevertheless known: a layer
 43 filled with voxels. Because cubes tile space, a layer of identical voxels does
 44 not approximate a homogeneous layer; it *is* one. Any difference between
 45 its reflection and transmission and those of the continuous layer is therefore
 46 error of the discretisation alone, with no modelling error and no shape error
 47 behind which to hide. A sphere, the usual test object for volume-integral and
 48 discrete-dipole methods [3], cannot give this: its error mixes the discretisation
 49 with the staircase approximation of a curved surface by cubes. The layer
 50 plays the part of the patch test in finite elements: reproducing a uniform
 51 medium correctly is necessary, though not sufficient, for a scheme to be
 52 trusted in a heterogeneous one. The experiment is then made harder in
 53 three steps, each with an exact solution of its own: a randomly stratified
 54 layer, in which every cell is homogeneous but the cells differ; a layer whose
 55 properties vary smoothly within every cell, which is where the basis for the
 56 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 57 surface, which carries the layer's prediction into three dimensions without a
 58 staircase. The coupling and the single site that the layer validates are the
 59 same in any medium; a laterally heterogeneous medium remains to be tested.

60 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges
 61 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 62 each cell therefore caps every voxel at second order in a medium that varies
 63 within the cells, however rich the basis of its field; and a contrast of higher
 64 degree than the field is wasted. (2) *The leading error is in closed form, for*
 65 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$

66 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the orders 2, 4 and 6 of
 67 $p = 0, 1$ and 2 hold for P, SV and SH waves at normal and oblique incidence.
 68 For the collocation scheme, the uniform-field voxel in its simplest form, the
 69 error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§7), and the
 70 opposite signs explain why no correction of the single site can improve both
 71 (§12). (3) *The error that remains is the single site's, and it is the projection*
 72 *error of the medium.* The single-site T -matrix sums the rescattering within a
 73 voxel to all orders and the coupling between voxels is exact, but the T -matrix
 74 is that of the voxel as represented, its contrast projected to degree r and its
 75 field held to degree p . The error of the term of second order in the contrast
 76 equals the *relative projection error* of the medium, the fraction of the squared
 77 norm of its profile that polynomials of degree $\min(p, r)$ on the cells cannot
 78 represent (Eq. (43)): exactly in the long-wave limit, and to within 1.3% for
 79 all nine pairs with $p, r \leq 2$ (§9). The wavefield basis must therefore carry
 80 what the medium imprints on the field, not only the wave, and the two bases
 81 must be matched. (4) *The error is known in advance.* The projection error
 82 is a property of the medium and the grid alone, so the cell size and the
 83 degree that a required accuracy needs are chosen before anything is solved.
 84 (5) *Where the medium is piecewise constant on the cells the projection error*
 85 *vanishes*, a constant suffices for the medium, and the degree of the field
 86 alone sets the order: in a randomly stratified model one first-moment voxel
 87 per model layer reaches a reflection accuracy of 10^{-6} , where the uniform-field
 88 voxel needs about forty (§9), and in the uniform layer it allows voxels 58 times
 89 larger in each direction (Appendix Appendix I). (6) *In three dimensions* the
 90 smoothly graded sphere converges at second order with the mean-only voxel,

91 as the layer predicts (§10), and at fourth order with the first-moment voxel
 92 (§11).

93 *What makes the experiment exact.* The answer above can be trusted only if
 94 nothing but the two bases is approximated. Three results, set against the
 95 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 96 The integrated Green tensor and the filtered coupled dipoles approximate the
 97 field of a cell; for cubes the source-cell average is exact, because the cube's
 98 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 99 at normal incidence the lattice kernel is in closed form with no lattice sum.
 100 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 101 coupled-dipole method, is here a derived object (§6): the moments of the
 102 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 103 without adding it by hand, the cube's departure from the sphere as a pure
 104 cubic anisotropy, and the closure's error against exact scattering by a sphere
 105 as a real $(ka)^2$ term in closed form. The self term the single site carries is the
 106 one the averaged kernel omits, so the discrete system is a collocation of the
 107 continuum equation. (c) *The elastic state.* A voxel carries a force (through
 108 its density contrast) and a moment (through its stiffness contrast), so its
 109 state has nine components, displacement and strain, the coupling is a 9×9
 110 tensor, and the single site's self-interaction is the elastic depolarisation of a
 111 cube, the discrete counterpart of the inclusion tensor of Eshelby [4]. Cubes
 112 serve all three: they tile space, so a uniform region cut into cubes is still that
 113 region, and their regular lattice makes the coupling a convolution, solved by
 114 fast transforms.

115 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
 116 small scatterers that interact through the background Green's function is the
 117 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics
 118 it became a numerical method as the coupled-dipole method, or discrete
 119 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
 120 dipole driven by the incident field and by the fields of every other dipole
 121 [7, 8]. The method is a collocation of the volume integral equation for the
 122 electric field, with the cell's self-interaction absorbed into its polarisability
 123 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns
 124 on two ingredients, and this paper takes up both for elasticity. The first is the
 125 polarisability. The Clausius–Mossotti value is the static response of a sphere;
 126 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that
 127 an infinite lattice propagates waves as the continuum does to a given order
 128 in kd ; Rahmani et al. [12] instead derive it from the exact static solution for
 129 the whole body, so that the method is exact in the long-wavelength limit.
 130 The second is the coupling. Integrating the Green tensor over the source
 131 cell rather than sampling it at the centre improves the accuracy, most of
 132 all for strong contrasts [13], and the filtered coupled dipoles of Piller and
 133 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace
 134 the point dipole by a source of finite extent whose spectrum is band-limited
 135 to the grid. The coupling summed over a lattice extends the method to
 136 periodic structures [16]. Its error converges at second order in the dipole
 137 spacing when shape errors are absent [3], and for a cube, the one shape
 138 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach
 139 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the

140 mathematical side, Costabel et al. [18] show that the method is a finite section
 141 of an infinite block Toeplitz matrix, stable for a large range of the contrast
 142 but not for all of it, and Costabel [19] that in a one-dimensional model it
 143 converges at second order in the interior, with a boundary layer in which
 144 the consistency error does not vanish. The layer studied here is the elastic
 145 counterpart of that cube: a body the grid represents exactly, so that the
 146 discretisation error can be isolated. What this paper adds is the closed form:
 147 for cells that tile space, the coupling that makes the lattice reproduce the
 148 continuum exactly, and the order and the constant of the error that remains,
 149 as functions of the two bases. In the terms used here the coupled-dipole
 150 method is the pair $p = r = 0$, and its refinements improve the polarisability
 151 and the coupling while leaving both bases constant in the cell.

152 *Its elastic counterparts.* The elastic problem has the same volume integral
 153 equation [20], and every cell-based solver of it is a coupled-dipole method
 154 whether or not it carries the name. Nestor [21] divided a two-and-a-half-
 155 dimensional medium into cells that interact through the background Green's
 156 tensor, each carrying a contrast renormalised by its own near field, with
 157 the multiple-scattering series summed. Kanaun and Levin [22] discretise the
 158 inclusion problem with Gaussian approximating functions whose matrix ele-
 159 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral
 160 equation on a grid by conjugate gradients with fast Fourier transforms, and
 161 Touhei [24] does the same for a half space with a generalised Fourier trans-
 162 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and
 163 inverse problems on the transition operator of the discretised Lippmann-
 164 Schwinger equation, the first in the acoustic approximation and the second

165 in arbitrary anisotropic elastic media with a nine-component state vector.

166 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and is
 167 ordered as the experiment is built. §§2–3 set out the discrete medium and the
 168 continuous layer it must reproduce; §4 shows why the point coupling fails,
 169 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
 170 single-site T -matrix as the first member of a Cartesian multipole hierarchy in
 171 closed form, and sets out what that gives (§6.7). §§7–8 take up the wavefield
 172 basis, §9 the medium basis and its projection error, §10 the sphere, and
 173 §11 the hierarchy as a voxel scheme. §12 discusses what the results imply
 174 and §13 concludes. Every result is derived symbolically in Mathematica and
 175 checked against an independent Python/Fortran implementation or an exact
 176 solution.

177 2. Setting

178 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 179 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 180 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 181 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 182 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 183 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

184 displacement followed by engineering strain. A voxel responds with a force
 185 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 186 state to them, and the 9×9 lattice Green’s tensor $\mathbf{K}(i, j)$ maps a source

187 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

188 with ψ^0 the incident state.

189 *Where the kernel comes from.* Write the medium as the background plus
 190 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 191 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 192 ting $\rho = \rho_0 + \Delta \rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta \lambda \delta_{bc} \delta_{de} + \Delta \mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 193 the contrast stiffness, and moving the contrast terms to the right leaves the
 194 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

195 a force from the density contrast, and the divergence of the contrast stress m ,
 196 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the
 197 background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
 198 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

199 the boundary term vanishing because m is confined to the scatterers, and the
 200 last step holding because $\mathbf{G}$ depends only on the separation. The moment
 201 therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when
 202 the moment is the contrast stress itself. With the moment tensor defined
 203 instead through its equivalent body force $-\partial_c [M_{bc} \delta(\mathbf{x} - \mathbf{y})]$, as in Aki and
 204 Richards [27], the same field reads $M_{bc} \partial G_{ab} / \partial y_c$, the source derivative, with

205 $M = -m$. The two are one convention written two ways; what matters is that
 206 the derivative in the kernel and the sign in the contrast operator are chosen
 207 together. The nine-component kernel follows (Fig. 1): its force columns are
 208 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 209 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
 210 those entries and its strain rows their receiver derivatives, symmetrised in
 211 the same way and doubled on the shear rows to give engineering strain. In
 212 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

213 where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$
 214 and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

215 *Averaging, and collocation.* Three operations in this paper are averages over
 216 a cell, and they are distinct. The *source-cell average* integrates the kernel
 217 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 218 of a uniform polarisation of the source cell. Every scheme in this paper uses
 219 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 220 is imposed: at the receiving cell's centre, with the internal field of each cell
 221 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 222 *average* imposes the equation in the mean over the receiving cell instead;
 223 with a uniform internal field that is the Galerkin scheme of degree zero,
 224 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 225 term that refers to the internal field and not to the test, and they differ only
 226 in the test function (a delta at the centre against the cell's indicator) and
 227 in the constant of their $O((kd)^2)$ error (Table 8). The *cell-mean contrast*,

228 finally, is an average of the material, not of the field: where the medium
 229 varies within a cell, the contrast is projected onto the cell’s basis, its mean
 230 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 231 of the coupled-dipole method, the point dipole is collocation with the point
 232 kernel, the integrated Green tensor adds the source-cell average, and the
 233 filtered coupled dipoles replace the point source by one of finite extent.

234 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 235 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 236 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 237 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 238 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

239 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 240 join the centres of the voxels in one plane (a square lattice of pitch d), and
 241 m is the separation of the two planes in units of d (Fig. 2a,b). The question
 242 is whether, and how fast, this chain converges to the continuous layer as
 243 $n \rightarrow \infty$ at fixed D .

244 *Implementations.* Every result is derived symbolically in Mathematica and
 245 compared with an independent numerical implementation or an exact so-
 246 lution. “The package” below is the Python/Fortran implementation of the
 247 voxel scheme (`cubic_scattering`): its cube T -matrix, its lattice kernels and
 248 its stratified propagator, which serve both as the object under test and, where
 249 validated, as a second implementation.

250 3. The continuous layer and its thin-layer expansion

251 At normal incidence a plane force drives a one-dimensional problem per
 252 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 253 Green's function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer's reflection and transmission
 254 follow from continuity of displacement and traction at its faces; they agree
 255 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 256 P and S (`ContinuumLimit_Reference.wl`).

257 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 258 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 259 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 260 cient expands as

$$R = \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

261 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 262 coefficient symbolically and the truncated series against the exact layer).
 263 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

264 to first order a thin layer is an interface across which the traction jumps by
 265 the excess inertia and the displacement by the excess compliance. At oblique
 266 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 267 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 268 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients

269 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 270 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 271 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 272 a single plane of voxels must reproduce order by order. A plane of voxels
 273 that carry Legendre moments of their internal field to degree p reproduces
 274 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 275 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 276 first-moment voxel all four terms shown.

277 **4. The specular sums of the point Green's tensor**

278 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 279 of every voxel of a plane m planes away. In that form it is a poor object to
 280 study. The Green's tensor decays slowly with distance, so the sum converges
 281 slowly, and it hides the one question that matters here: which part of it is
 282 the continuum, and which part is the discretisation. The Poisson summation
 283 formula [28] answers both at once, and is the standard tool for lattice sums
 284 of wave fields [29]. For a function f on the plane and the square lattice of
 285 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

286 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 287 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).
 288 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

289 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 290 between planes a distance z apart. Each term has a physical meaning. Trans-
 291 formed laterally, the field of a plane of sources is a superposition of plane
 292 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 293 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 294 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 295 far beyond the wavenumber of the field when the voxels are small compared
 296 with the wavelength, so it is evanescent and decays away from the plane as
 297 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 298 the lattice sum exactly into the continuum and a discretisation correction,
 299 and the correction converges after a few orders.

300 *The local term that survives.* In the strain-from-moment block the evanescent
 301 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 302 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 303 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 304 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 305 planes in slab order, the static lattice sum of the point kernel is exactly
 306 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 307 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 308 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 309 reproduces the continuum only if that remainder is cancelled by the voxel's
 310 own self term.

311 5. The tiling identity

312 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 313 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 314 Each Poisson order in Eq. (10) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

315 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 316 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 317 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube’s indicator χ : a partition of unity (Fig. 4).
 318 Two exact statements follow.

319 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

320 the continuum’s plane-wave term times the vertical form factor. The pack-
 321 age’s cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 322 `AveragedSums.wl`).

323 *Within a plane.* The sum over *all* cells of the plane, the voxel’s own included,
 324 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

325 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 326 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 327 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 328 three normal strain-from-moment entries, whose local weights $-1/M_P$ and

329 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 330 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 331 with every cell integral evaluated analytically (below).

332 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 333 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

334 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 335 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 336 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 337 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 338 and the quadrature order are raised; Eq. (14) is now the package’s kernel at
 339 normal incidence.

340 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 341 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 342 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 343 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 344 which never contain the origin, and there the two-dimensional antiderivative
 345 in the face coordinates is continuous, so the face integral is a sum over the
 346 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 347 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 348 *not* usable for repeated derivatives: its arctangent terms jump across the
 349 coordinate planes through the origin, and differentiating twice there discards
 350 exactly that term.) The self term so obtained reproduces the classical closed
 351 form of the cube’s depolarisation [30, 31] to machine precision.

352 6. The single site: the cube T -matrix from moment integrals

353 Every error of a voxel scheme is made at the single site (§12), so the single
354 site is derived here rather than imported. The classical route imports it. For
355 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
356 uniform load uniform, with a depolarisation tensor in closed form [4, 32, 33];
357 its dynamic generalisation is known for spheres and ellipsoids [34, 35, 36].
358 For a cube or a general polyhedron the static field of a uniform eigenstrain
359 is also in closed form [30, 37, 31], and it has been computed for a cuboid in
360 an anisotropic matrix [38], but the strain it produces is not uniform, so there
361 is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors
362 for polyhedral inclusions with polynomial eigenstrains have since been given
363 [39], the dynamic inclusion problem has been formulated for shapes other
364 than the ellipsoid, cubes among them [40], and the Green’s tensor has been
365 integrated analytically over a cuboid for the coupled-dipole method, to an
366 error of $O((kd)^4)$ [41]; these are the static and electromagnetic counterparts of
367 individual moments below. In scattering theory the volume integral equation
368 of an elastic inclusion [20] is closed in the same spirit by a self-consistent
369 or effective-field approximation [42, 43]: a first-order (Born) scatterer [44]
370 renormalised by the geometric series of its own near field, the step that turns
371 a contrast into an effective contrast. Nestor [21] introduced the single site
372 in this form, summing the series of the cell’s self integral, with the singular
373 near field evaluated over an infinitesimal cell. The moments below evaluate
374 the same self integral over the finite cube, as a distribution. The elastic T -
375 matrix of Waterman [45] represents a scatterer in spherical multipoles, the
376 natural basis for a sphere; a voxel in a lattice needs one compatible with the

377 lattice, and the moments over the cube provide it. The construction below
 378 makes the approximation of a voxel explicit rather than implicit: it derives
 379 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 380 over the cube, with the uniform-strain closure as the first truncation of a
 381 hierarchy.

382 *A Cartesian multipole hierarchy.* We call the construction the Cartesian mul-
 383 tipole hierarchy (Fig. 5). The contrast acting on the voxel's field is a source:
 384 the density contrast a body force $\omega^2 \Delta \rho \mathbf{u}$, the stiffness contrast a stress
 385 $\delta \mathbf{c} : \boldsymbol{\varepsilon}$. Each voxel radiates through the Cartesian multipole moments of
 386 these sources over the cube, a force monopole and a stress dipole from the
 387 uniform field (the cell of the coupled-dipole method) and higher multipoles
 388 from the field's gradients, and receives the field through its gradients at its
 389 centre. The name needs one caution. A multipole expansion usually expands
 390 the kernel or the field about a centre and truncates it, which holds only for
 391 well-separated cells or outside a circumscribing sphere: in the fast multipole
 392 method [46], whose elastodynamic form leaves adjacent cells to direct inte-
 393 gration [47], in the spherical-wave T -matrix of Waterman [45], and in the
 394 force-dipole description of a point defect, an expansion of the Green's func-
 395 tion far from it [48]. Here the Green's tensor is never expanded. Only the
 396 source is resolved into Cartesian moments, and the integral over the source
 397 cube is exact at every separation, so there is no radius of convergence, and
 398 touching voxels and the self term, a distribution in closed form, are included.
 399 The source side is shared with the multipole extensions of the coupled-dipole
 400 method in optics [49, 50] and with Cartesian multipole decompositions of a
 401 source [51]. The word hierarchy has its sense in kinetic theory and cosmology

402 [52]: a chain of coupled moment equations closed by truncation at a chosen
 403 order, whose truncation error is here in closed form.

404 *The multipole orders.* A hierarchy of degree ℓ holds the force density $\omega^2 \Delta \rho \mathbf{u}$
 405 to degree ℓ and the stress $\delta \mathbf{c} : \boldsymbol{\varepsilon}$ to degree $\ell - 1$. A stress moment of degree n
 406 acts on the medium through its divergence, as a force multipole of order $n + 1$,
 407 so both channels carry the force multipoles through order ℓ (Table 1). Here
 408 a multipole is the primitive Cartesian moment, which keeps its traces, rather
 409 than the irreducible one [51]. What sets the order of convergence is each
 410 channel's first omitted source term, counted in its own degree (§6.4): fourth
 411 order needs the force quadrupole for a density contrast, but the second-degree
 412 stress moments, a force octupole, for a stiffness contrast.

Table 1: The force multipoles a hierarchy of degree ℓ carries, and the order of convergence it reaches for each contrast on the layer (Table 3).

ℓ	density channel	stiffness channel	order reached	
	(force moments)	(stress moments)	density	stiffness
1	monopole, dipole	dipole	2	2
2	to the quadrupole	dipole, quadrupole	4	2
3	to the octupole	to the octupole	4	4

413 *Its origin in the equivalent inclusion method.* In its static form the con-
 414 struction is the equivalent inclusion method with polynomial eigenstrains
 415 of Moschovidis and Mura [53]: the eigenstrain is expanded about the centre
 416 of the inhomogeneity, the consistency equation is expanded there in a Taylor
 417 series, and the coefficient of each monomial is equated, which amounts to

418 imposing the Lippmann–Schwinger equation and its derivatives at the cen-
 419 tre [54, §2.2]. Johnson et al. [55, 56] applied an expansion about the origin
 420 to a cuboidal inhomogeneity, and Wu et al. [39] expanded the eigenstrain
 421 of a polyhedron to second degree about its centroid; both report that the
 422 expansion about the centre does not follow the field near the edges, corners
 423 or vertices. Fu and Mura [34] extended the method to time-harmonic waves
 424 for an ellipsoid, with a second unknown for the density mismatch, and in the
 425 cases they report evaluated it with uniform eigenstrains. Brisard et al. [54]
 426 replaced the Taylor form by a Galerkin form of the same equation, because
 427 the Taylor form need not give an invertible system and a higher degree need
 428 not improve it. What is added here, to our knowledge, is the elastodynamic
 429 hierarchy, with density and stiffness contrasts together, to any degree and for
 430 a body that is not an ellipsoid; its moments as distributions in closed form;
 431 and the structure and the error of its truncations, tested degree by degree
 432 against the exact solution for a sphere. The moments here are integrals of
 433 the Green’s tensor against monomials over the body. They are not moments
 434 in the sense of the method of moments [57], of which the hierarchy is the
 435 instance with monomials about the centre as expansion functions and the δ
 436 function and its derivatives at the centre as testing functions; and the exact
 437 static response of an inclusion of any shape to polynomial loads is described
 438 by the polarisation and elastic moment tensors of Ammari and Kang [58].

439 *6.1. The closed hierarchy*

440 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 441 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. The hierarchy is stated below for any degree; its third degree
 442 is written out first, being the lowest at which the stiffness channel converges

443 at fourth order (§6.4). Expand the internal displacement about the centre to
 444 third degree and project the Lippmann–Schwinger equation onto 1, x_p , x_px_q
 445 and $x_px_qx_v$. The unknowns are $u_i(0)$, $\partial_p u_i(0)$, $\partial_p \partial_q u_i(0)$ and $\partial_p \partial_q \partial_v u_i(0)$, and
 446 the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$\begin{aligned} & (\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j \\ & + (\frac{1}{6} \omega^2 \Delta \rho \mathbf{B}_{ij,p}^{rst} - \frac{1}{2} \mathbf{P}_{in,pk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \end{aligned} \quad (16)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

$$\begin{aligned} & (\omega^2 \Delta \rho \mathbf{Q}_{ij,pqv}^r - \mathbf{X}_{in,pqvk} \delta c_{nkrj}) \partial_r u_j \\ & + (\delta_{pr} \delta_{qs} \delta_{vt} \delta_{ij} + \frac{1}{6} \omega^2 \Delta \rho \mathbf{J}_{ij,pqv}^{rst} - \frac{1}{2} \mathbf{Z}_{in,pqvk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \end{aligned} \quad (18)$$

447 all fields evaluated at the centre. Dropping the third gradients, that is
 448 Eq. (18) and the second line of Eq. (16), leaves the system of second de-
 449 gree; dropping the second gradients as well, that is Eq. (17) and the $\partial \partial u$
 450 terms of Eq. (15), leaves the uniform-strain closure. Equations (15) and (17)
 451 are the same at second and third degree, because every term the third de-
 452 gree would add to them vanishes by parity (§6.3). The coefficients are ten

453 moments of the background Green's tensor $\mathbf{G}$ over the voxel

$$\begin{aligned}
\mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\
\mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \\
\mathbf{B}_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{J}_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{X}_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} \, dV, \\
\mathbf{Z}_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s \, dV,
\end{aligned} \tag{19}$$

454 the first six being those of the second degree and the last four those the third
455 gradients add. The derivatives are taken with respect to the source point,
456 $\partial' = -\partial$. That sign reaches the moments with an odd number of derivatives,
457 $\mathbf{N}$, $\mathbf{Q}$, $\mathbf{B}$ and $\mathbf{J}$, and no symmetry, parity or scaling check can detect it, so it
458 is fixed by convention. The moments are graded by (D, W) , the number of
459 derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$, $\mathbf{M}(2, 0)$,
460 $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$, and $\mathbf{B}(1, 3)$, $\mathbf{J}(3, 3)$, $\mathbf{X}(4, 0)$, $\mathbf{Z}(4, 2)$.

461 *The hierarchy at any degree.* Nothing in the construction is special to the
462 third degree. Let the internal displacement be expanded to degree ℓ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \tag{20}$$

463 the sum running over ordered sets of indices $W = (w_1, \dots, w_m)$, and let the
464 strain be its derivative. Write the moments with any number of derivatives
465 and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \tag{21}$$

466 the derivatives being those of G_{in} as a function of its argument and the pairing
467 distributional (§6.2). The ten moments of Eq. (19) are these $\mathbf{g}$ themselves:

they are the integrals of $\partial'_D G_{in}(\mathbf{0} - \mathbf{x})$, and the factor $(-1)^{|D|}$ of the source derivative cancels the parity $(-1)^{|D|}$ of $\partial_D G_{in}$, so that, for example, $\mathbf{N}_{in,k}^r = \mathbf{g}_{in}[k; r]$ and $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$. Beyond the third degree the moments are written as $\mathbf{g}$ only. Imposing the Lippmann–Schwinger equation and its derivatives to order ℓ at the centre gives, for every set of derivative indices P with $|P| \leq \ell$,

$$U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \quad (22)$$

Equations (15)–(18) are the case $\ell = 3$, with the moments of Eq. (19) put for the $\mathbf{g}$; their truncations above are $\ell = 2$ and the uniform-strain closure $\ell = 1$. With the field rather than the eigenstrain as the unknown, the moments $\mathbf{g}[D; W]$ play the part of the polynomial Eshelby tensors of the equivalent inclusion method [33, Ch. 4], which Brisard et al. [54] call influence pseudotensors, here for the elastodynamic kernel and for the density channel as well. For an ellipsoid the static self term of a polynomial eigenstrain is itself a polynomial of the same degree, so the expansion about the centre is exact in the self term and approximate only in the interaction with other bodies [54]; for a cube it is approximate in both. The cube is centrosymmetric, so a moment vanishes unless $|D| + |W|$ is even, and the system needs the grades

$$\begin{aligned} (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\ (D, W) &= (m+1, W), \quad W \leq \ell-1 && \text{(stiffness),} \end{aligned} \quad m = 0, \dots, \ell, \quad D+W \text{ even.} \quad (23)$$

Table 2 lists what each degree adds. The unknowns number $3 \binom{\ell+3}{3}$: 12, 30, 60 for $\ell = 1, 2, 3$.

Table 2: The grades (D, W) of moments, D derivatives on the Green's tensor and a weight of degree W , that the hierarchy of degree ℓ needs beyond those of degree $\ell - 1$.

ℓ	new grades	moments in all
1	(0, 0), (1, 1), (2, 0)	3
2	(0, 2), (2, 2), (3, 1)	6: G, K, N, M, P, Q
3	(1, 3), (3, 3), (4, 0), (4, 2)	10: with B, J, X, Z
4	(0, 4), (2, 4), (4, 4), (5, 1), (5, 3)	15
5	(1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4)	21

487 *Two independent systems.* In either sum of Eq. (22) the unknown $U_{i;P}$ is
488 coupled to $U_{j;W}$, or to $U_{j;rW}$, through a moment of grade $(|P|, |W|)$ or $(|P| +$
489 $1, |W|)$, and on a centrosymmetric body that moment vanishes unless $|P|$ and
490 the degree of the unknown it multiplies have the same parity. The hierarchy
491 of any degree therefore separates exactly, at every frequency and contrast,
492 into two systems: one for the even gradients $u, \partial\partial u, \dots$ and one for the odd
493 gradients $\partial u, \partial\partial\partial u, \dots$. Raising the degree from odd to even adds a block
494 to the even system alone, and from even to odd a block to the odd system
495 alone. At $\ell = 3$ the even system is Eqs. (15) and (17), unchanged from $\ell = 2$,
496 and the odd system is Eqs. (16) and (18). Of the sixty unknowns, twenty-one
497 belong to the even system and thirty-nine to the odd. The leading radiation
498 of a density contrast, $\int_V \Delta \rho u \, dV$, is drawn from the even system, and that of
499 a stiffness contrast, $\int_V \delta c : \nabla u \, dV$, from the odd one. The separation is thus
500 the structure behind the result of §6.4, that the blocks improve the density
501 and the stiffness channels alternately. At first degree it is implicit in the
502 dynamic equivalent inclusion method for an ellipsoid, where the force and

503 the strain systems uncouple at the centre [34]; here it holds at every degree,
 504 for any centrosymmetric body.

505 6.2. The moments as distributions

506 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (24)$$

507 [59, 27]. Expanding g_c in powers of r reduces every moment in Eq. (19) to a
 508 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (25)$$

509 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 510 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 511 gral depends on the shape of the region excluded around the origin. This is
 512 the source of the δ -function term of the inclusion problem, which a numerical
 513 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 514 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 515 are distributional from the start, and each is moved onto the voxel's faces by
 516 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (26)$$

517 The surface term is evaluated on the faces, a distance h from the origin, where
 518 every derivative of r^m is smooth. The recursion lowers D and W together
 519 and ends on convergent integrals. The face integrals are done in closed form
 520 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (27)$$

521 which the face route satisfies exactly and a spherical excision returns as
 522 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
 523 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 524 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 525 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 526 Because every power of r is paired this way, the dynamic moments keep the
 527 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
 528 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 529 those terms, at a real $O((kh)^2)$ cost (§6.6).

530 *Every grade, in closed form on four constants.* Closed forms for monomial-
 531 weighted integrals of $1/r$ over bricks are known [60], as are closed forms for
 532 the potential and its first two derivatives over polyhedra with polynomial
 533 density, from the gravity literature [61]. The engine used here differs in
 534 taking derivatives of any order as distributions, in treating the kernel r as
 535 well as $1/r$, and in the even-power terms of the dynamic series. The recursion
 536 (26) does not depend on the grade, so the same engine gives every scalar
 537 moment of Table 2. Each is evaluated once and stored. The scalar moments
 538 of every grade through $\ell = 5$, with up to six derivatives and weights to
 539 the fifth degree, have been evaluated for the kernels r^{-1} and r : 4224 closed
 540 forms. (The second term of Eq. (24) places two further derivatives on r ;
 541 those moments are stored through $\ell = 3$ and are evaluated numerically, by
 542 the second route below, beyond it.) Every one of them is a power of the
 543 voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (28)$$

544 The integration returns many of them in other forms, with $\log(\sqrt{3} - 1)$,
545 $\log(1 + \sqrt{3})$, inverse hyperbolic tangents and logarithms of powers of two,
546 which are equal in value but not canonical: two equal moments can then differ
547 in appearance, and sums of large logarithms that cancel lose digits. Each
548 stored form is therefore reduced to Eq. (28) by an integer-relation search at
549 400 digits, with $\log 2$ and $\log 3$ admitted to the search, and the reduction
550 is accepted only if it agrees with the stored form to 1200 digits. All 4224
551 reduce, and the coefficients of $\log 2$ and $\log 3$ are zero in every one.

552 Every grade is gated by two checks whose answers do not depend on
553 the engine, the Laplacian sum rules of Eq. (A.1), which test the δ function
554 of the inclusion problem and its derivatives where a quadrature fails, and
555 confirmed by a second route that shares nothing with the faces of the cube,
556 to 1.1×10^{-12} or better (Appendix Appendix A).

557 *6.3. Parity, and the nine-component block*

558 The cube is centrosymmetric. A moment that couples a field quantity of
559 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
560 $\partial\partial u$ have degrees 0, 1 and 2, so at second degree Eq. (16) contains the first
561 gradient alone: ∂u is determined by its own block and couples to neither u
562 nor $\partial\partial u$. At any degree it couples to nothing of even degree; from $\ell = 3$
563 it couples to $\partial\partial\partial u$ (the second line of Eq. (16)). The leading scattering by
564 a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u .
565 The leading far field of the stiffness contrast is therefore fixed by Eq. (16),
566 and *no treatment of the second-gradient block (17) can change it*. What
567 Eq. (17) contributes reaches the far field only through the $O((kh)^2)$ coupling
568 in Eq. (15). That is the order, and the origin (the variation of the field across

the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes, as does the hierarchy itself once its third-gradient block is kept (§6.4).

The symmetry of the cube. The cube is unchanged by 48 operations: the 24 rotations that carry it into itself, each with or without the inversion $\mathbf{x} \rightarrow -\mathbf{x}$. They form the group O_h . Any operator built from the cube alone commutes with all 48, and group theory then says how far it can be simplified [62, 63]. The quantities it acts on split into irreducible representations, the smallest sets of components that the symmetry operations mix only among themselves. Such an operator has no matrix elements between different representations, and on a representation that occurs once it is a multiple of the identity (Schur’s lemma). It is then diagonal, with one eigenvalue for each representation, however many components that representation has. The labels are the standard ones of crystal physics. A , E and T are representations of dimension one, two and three, and the subscript g (from the German *gerade*, even) marks those that are even under inversion, as a strain is. For the nine components of a displacement gradient, A_{1g} is the dilatation, E_g the two traceless combinations of the diagonal strains, T_{2g} the three off-diagonal shears, and T_{1g} the three rotations, the antisymmetric part of the gradient. A stiffness contrast acts on strain only, so on T_{1g} the operator below is the identity. A sphere is invariant under every rotation, not only the cube’s, and there E_g and T_{2g} join into a single five-dimensional shear channel. The cube splits that channel into two and three dimensions, which is why its two shear eigenvalues are averaged with the weights 2 and 3 in Eq. (32) below.

With $\Delta\rho = 0$, Eq. (16) at second degree is the 9×9 operator $\delta_{pr}\delta_{ij} -$

594 $M_{in,pk} \delta c_{nkrj}$, built from M alone. Under O_h the nine gradient components
 595 decompose as $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis.
 596 Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

597 The density channel is the displacement block of Eq. (15) with the second
 598 gradients dropped, an amplification $(1 - \omega^2 \Delta\rho \mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 599 of $\mathbf{G}$.

600 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 601 tional channel $1 + \Delta K / (\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 602 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 603 latational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves
 604 M only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 605 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 606 the shape. The cube's shear channels differ from each other. Their isotropic
 607 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

608 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 609 order is therefore a pure cubic anisotropy. It splits the deviatoric channel

about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g , and the diagonal shear is the more depolarised. At the background of §7 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is 0.4960.

Where Eshelby enters. Equations (29)–(31) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no δ term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (22) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree; Table 5 measures what each degree leaves. A voxel in a uniform layer has no such concentrations: its neighbours have the same properties, so its internal field is the continuum's, which is smooth. That is why a low-degree internal field suffices in a lattice (§§7–8) although it is only an approximation for an isolated cube.

Against the exact sphere. These statements are tested against the exact elastic Mie solution, with no Eshelby formula in between (Appendix Appendix F). Applied to a ball, the construction's dilatational and shear channels equal Mie's static concentrations to 2×10^{-10} , and the cube's dilatational channel and isotropic shear average meet the same numbers; taken as a principal value the moment has no depolarisation at all, so the whole

static response is the δ term. At finite frequency the closure departs from
Mie only by a real term of order $(ka)^2$ at first order in the contrast, given
in closed form in Eq. (F.1); the radiation damping is exact. That term is
the uniform-field ansatz's own error, the order the hierarchy assigns to the
omitted second gradients.

6.4. What each block of the hierarchy buys

A hierarchy invites the question of what its next block is worth. The
answer has a simple form, and it is not the one a count of unknowns suggests:
the blocks help in pairs.

The experiment. The layer of §7 at normal incidence is cut into n cells, and
in each the hierarchy of degree ℓ is imposed: the displacement is its Taylor
polynomial of degree ℓ about the centre, the strain the derivative of that, and
the equation and its first ℓ derivatives hold at the centre, with the δ terms
kept. In one dimension the moments are elementary and the exact layer is
known, so the order of each truncation can be measured for a density contrast,
a modulus contrast and both. The schemes are computed symbolically to 60
digits for $\ell = 1$ to 6 and, to $\ell = 3$, independently in double precision.

The prediction. A cell radiates the integral of its source against the kernel.
If the source is held to a polynomial of degree d about the centre, the first
power it omits is $d + 1$. When $d + 1$ is odd, that term integrates to zero
against the even part of the kernel and the error is two orders smaller; when
 $d + 1$ is even it does not. The density source is the displacement, of degree ℓ ;
the stiffness source is the strain, of degree $\ell - 1$. So a density contrast gains

an order pair at every even ℓ , and a stiffness contrast at every odd ℓ . This is the parity of §6.3, read as an order of convergence.

Table 3: The order of convergence of the hierarchy of degree ℓ on the layer, measured from 16 to 32 cells in reflection and in transmission at $\omega = 300 \text{ rad s}^{-1}$, and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

	$\ell = 1$	2	3	4	5	6
density contrast only	2	4	4	6	6	8
modulus contrast only	2	2	4	4	6	6
both	2	2	4	4	6	6
error, both, $n = 16$	5.4×10^{-6}	1.4×10^{-6}	5.8×10^{-12}	9.5×10^{-13}	2.6×10^{-18}	3.1×10^{-19}

The result. Table 3: all thirty-six orders are as predicted. The second gradients raise the density channel to fourth order and leave the stiffness channel at second, with a constant three to four times smaller in reflection; the third gradients bring the stiffness channel to fourth order; and so on in pairs. With both contrasts, fourth order therefore needs $\ell = 3$ and sixth order $\ell = 5$. The truncation of Eqs. (15)–(18) at second gradients is one block short of the next order for a stiffness contrast, which is why it improves the uniform-strain closure by a constant and not by an order.

The error in closed form. At first order in the contrast a cell whose source is held to its Taylor polynomial of degree d returns the exact term times $1 + E_d$,

670 where, with h the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (33)$$

671 R and T standing for reflection and transmission, with $d = \ell$ for the den-
 672 sity term and $d = \ell - 1$ for the stiffness term. It follows from the series
 673 of $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$ with P_d the Taylor polynomial of e^{ikhx} , and it is
 674 confirmed symbolically for $d = 0$ to 6 and against the scheme on one cell,
 675 at a contrast of 10^{-20} and $kh = 0.03$, to 0.5% in all twelve cases. For
 676 $d = 0$ it is $(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission: the con-
 677 stants of the collocation scheme, Eq. (36), which are thus the first rung
 678 of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_`
 679 `layer_taylor_hierarchy.py`).

680 6.5. The single site against a sphere and a cube

681 The hierarchy is a statement about one scatterer, so it is tested first on
 682 one scatterer. The assembly of Eq. (22), with the full tensor moments in
 683 their dynamic form (the series of e^{ikr}/r taken to r^{11}), is the same for any
 684 centrosymmetric body; only the scalar moments change.

685 *A sphere.* For a ball the scalar moments are elementary, by the surface route
 686 of Eq. (26) on $|\mathbf{r}| = a$, and the exact answer is known. A homogeneous
 687 sphere of radius $a = 10$ m with the contrast of §7 is illuminated by a P plane
 688 wave; the hierarchy's polynomial source is radiated to nine scattering angles

689 between 11° and 169° , and the error is the largest component of the difference
690 from the exact displacement [64, 65] over the largest exact component. The
691 prediction, from the pairs of §6.4 with kh replaced by ka , is an error of
692 $O((ka)^2)$ for $\ell = 1$ and 2 and of $O((ka)^4)$ for $\ell = 3$, and $O((ka)^4)$ already
at $\ell = 2$ for a density contrast alone. Table 4 bears the prediction out. At

Table 4: One homogeneous sphere by the hierarchy of degree ℓ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

contrast	ℓ (unknowns)	$k_{Sa} = 0.4$	0.2	0.1	order	predicted
density only	1 (12)	6.4×10^{-3}	1.6×10^{-3}	3.9×10^{-4}	2.0	2
	2 (30)	4.8×10^{-5}	2.6×10^{-6}	1.8×10^{-7}	3.9	4
	3 (60)	1.5×10^{-5}	9.5×10^{-7}	6.9×10^{-8}	3.8	4
moduli only	1	1.9×10^{-2}	5.0×10^{-3}	1.1×10^{-3}	2.3	2
	2	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	2.0	2
	3	6.2×10^{-5}	4.1×10^{-6}	2.0×10^{-7}	4.3	4
both	1	1.4×10^{-2}	3.1×10^{-3}	7.2×10^{-4}	2.1	2
	2	3.2×10^{-3}	6.9×10^{-4}	1.8×10^{-4}	1.9	2
	3	4.6×10^{-5}	2.5×10^{-6}	1.4×10^{-7}	4.2	4

693
694 $k_{Sa} = 0.4$ the error of the uniform-strain closure is 1.4%, that of the second-
695 gradient system 0.32%, and that of the third-gradient system 0.0046%: three
696 hundred times below the closure. The systems are well conditioned, the sixty
697 equations of $\ell = 3$ having a condition number of 5.9 (`scripts/measure_`
698 `ball_gradient_hierarchy.py`).

699 *A cube.* The same assembly with the cube’s moments gives the single site of
700 a voxel. No exact solution exists for a cube, so the reference is the same cube,
701 of half-width $h = 10$ m, solved by the Legendre voxels of §8 on refined grids,
702 a cube being the one body such a grid represents exactly. Solutions with first
703 and second moments on 2^3 and 3^3 cells differ among themselves by at most
 2×10^{-5} , which is the accuracy of the reference. Table 5 shows what distin-

Table 5: One homogeneous cube of half-width h by the hierarchy of degree ℓ : relative far-field error against the converged reference.

ℓ (unknowns)	$k_S h = 0.4$	0.2	0.1	0.05
1 (12)	2.1×10^{-2}	5.4×10^{-3}	2.8×10^{-3}	2.7×10^{-3}
2 (30)	3.2×10^{-3}	1.6×10^{-3}	1.9×10^{-3}	2.4×10^{-3}
3 (60)	3.5×10^{-4}	3.4×10^{-4}	2.9×10^{-4}	3.1×10^{-4}

704
705 guishes a cube from a sphere. Each truncation has a floor that does not fall
706 with the frequency, because the internal field of an isolated cube is not a poly-
707 nomial even statically: it concentrates at the edges and corners, as the static
708 expansions about the centre of Johnson et al. [56] and Wu et al. [39] found for
709 the cuboid and for polyhedra. The floor is 2.7×10^{-3} for the uniform-strain
710 closure and nearly the same with second gradients, as §6.3 requires, the static
711 stress dipole being fixed by the first-gradient block and coupled next to the
712 third. The third-gradient block lowers it ninefold, to 3×10^{-4} . Above the
713 floor the error falls with the frequency as on the sphere: at $k_S h = 0.4$ the
714 third-gradient system is sixty times more accurate than the closure. The dip
715 of the second-gradient row at $k_S h = 0.2$ is a cancellation between its static
716 and dynamic terms, which have opposite signs. The condition numbers lie be-

717 tween 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These
 718 results come from one implementation, checked against the exact sphere and
 719 against the converged cube; the moments it uses are those confirmed by two
 720 routes in §6.2.

721 6.6. The collocation closure

722 The collocation closure takes the internal field of a voxel as uniform, the
 723 truncation of §6.3 for the density and stiffness channels together. The voxel’s
 724 own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 725 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 726 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (34)$$

727 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 728 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (35)$$

729 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 730 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 731 so Eq. (35) is precisely the collocation of the continuum Lippmann–Schwinger
 732 equation with a strain piecewise constant over voxels. It does not contain
 733 $\mathbf{S}_{\text{self}}$ at all.

734 The package’s cube T -matrix is this closure statically: both blocks agree
 735 with Eq. (34) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 736 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about $0.25(kh)^2$

737 through its far-field form-factor corrections. Its self integrals are exact in the
738 static and radiation parts but omit the real even-power terms of the Green's
739 tensor's expansion ($k^2 r$, r^3 , $\dots$), an error of $0.4\text{--}0.6 (kh)^2$ against the analytic
740 self term, whose series agrees with the closed-form elastodynamic tensor to
741 10^{-14} . By Eq. (35) that error cannot enter the closure scheme.

742 6.7. *What a closed-form single-site T -matrix gives*

743 The Cartesian multipole hierarchy costs more to set up than the alter-
744 natives: moments of the Green's tensor to fourth derivatives, evaluated as
745 distributions and reduced to closed form. What that buys follows from the
746 Lippmann–Schwinger equation of one body rather than from a model fitted
747 to it. *One derivation.* The static depolarisation, its dynamic corrections,
748 the radiation damping and every higher block come from that one equation
749 and one engine (§6.2), so they cannot be inconsistent with one another, and
750 the static, spherical and low-frequency limits are results of the construction
751 rather than conditions imposed on it. *Nothing is fitted.* Every coefficient
752 of Eq. (22) is a moment of the background Green's tensor over the body:
753 there is no polarisability to choose and no depolarisation factor to calibrate,
754 where other routes import the single site from the ellipsoid [4, 33] or close
755 the volume integral equation with an effective field [42, 43]. As in the equiva-
756 lent inclusion method [53, 34], the uniform-strain closure is the first member
757 (§6.3); what the hierarchy adds is the closed form at finite frequency for a
758 body that is not an ellipsoid, with the error of each truncation known. *The*
759 *singular self-interaction is exact:* the δ -function content that a quadrature
760 misses is part of the moments from the start, and the sum rule (27) and the
761 traces (A.1) hold at every grade (§6.2). *The exact limits are built in:* the

762 cube's dilatational channel and its isotropic shear average equal the sphere's,
 763 confirmed against the exact Mie solution to 2×10^{-10} and 10^{-8} (§6.3), and
 764 at finite frequency the closure departs from Mie only through a real term
 765 of order $(ka)^2$, the radiation damping agreeing exactly in all three channels
 766 (Eq. (F.1)). *The error is stated before anything is solved:* a truncation is
 767 a statement of which blocks are omitted; the orders come in the predicted
 768 parity pairs, all thirty-six on the layer and all nine on the sphere (Tables 3
 769 and 4), and the leading error factor is in closed form, Eq. (33). *It improves*
 770 *block by block, on fixed constants:* raising the degree adds moments, con-
 771 firmed by a second route that shares nothing with the first (§6.2), and no
 772 quadrature; against the exact sphere at $k_{Sa} = 0.4$ the error falls from 1.4%
 773 for the uniform-strain closure to 0.0046% with third gradients, from a sys-
 774 tem of sixty equations with a condition number of 5.9 (Table 4). That each
 775 degree improves on the one below is shown here for one sphere and one cube,
 776 not in general: in the Taylor form of the equivalent inclusion method, for
 777 interacting inhomogeneities, a higher degree need not improve the solution
 778 and the system need not be invertible [66, 54]. *It says what a body's shape*
 779 *does:* at the order of the uniform-strain closure a cube is a sphere plus a pure
 780 cubic anisotropy, which splits the deviatoric channel about the sphere's value
 781 in the ratio 3 : 2 (Eq. (32)). *It describes a scatterer, not only a cell:* the as-
 782 sembly is the same for every centrosymmetric body, only the scalar moments
 783 changing (§6.5), and the same moments are the self block of a voxel scheme
 784 (§11).

785 *What it does not give.* A polynomial about the centre cannot follow the
 786 edge and corner concentrations of an isolated cube's internal field, so each

787 truncation there has a floor that does not fall with the frequency (3×10^{-4}
788 with third gradients; Table 5). And in a lattice the Legendre voxel of §8 is the
789 more economical in unknowns: both are fourth order on a smoothly graded
790 sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel
791 with thirty-six (§11). The hierarchy’s case is not the number of unknowns;
792 it is that it yields the single site itself, in closed form, with its error known.

793 **7. Convergence, and the closed-form error**

794 *The experiment.* Every convergence test of this paper is the same experiment,
795 varied in one respect at a time, and it is set out here once.

- 796 1. *The medium.* The background has $\alpha = 5000$ and $\beta = 3000 \text{ m s}^{-1}$ and
797 $\rho = 2500 \text{ kg m}^{-3}$. The layer occupies $0 < z < D$ with $D = 2 \text{ m}$,
798 and differs from the background by $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$ and
799 $\Delta\rho = 100 \text{ kg m}^{-3}$: 6.4% in the P modulus, 4.4% in the shear modulus
800 and 4% in the density.
- 801 2. *The source.* A plane force at $z_s = -12 \text{ m}$, above the layer, radiates a
802 plane wave towards it. The incident field in the layer is then known in
803 closed form, and it is smooth there because the source is outside.
- 804 3. *What is measured.* The scattered displacement, total less incident, at
805 two observers outside the layer: at $z = -1 \text{ m}$ above it, where it is the
806 reflected wave, and at $z = 4 \text{ m}$ below it, where it is the change the layer
807 makes to the transmitted wave. Outside the layer the scattered field is
808 a single outgoing wave, so one observer on each side holds everything;
809 the two are reported separately, as reflection / transmission.

- 810 4. *The exact answer.* The reflection and transmission of the layer from
811 continuity of displacement and traction at its two faces (§3); for a
812 layer whose properties vary with depth, from the differential equation
813 integrated to 30 digits.
- 814 5. *The error.* $e = |u_{\text{scheme}} - u_{\text{exact}}|/|u_{\text{exact}}|$ at each observer: relative to the
815 scattered field, not to the total, so that an error of 10^{-6} is a millionth
816 of what the layer scatters.
- 817 6. *The refinement.* The layer is cut into n planes of voxels of side $d =$
818 D/n , with n from 1 to 32, at $\omega = 60, 300$ and 600 rad s^{-1} , for which
819 $k_P D = 0.024, 0.12$ and 0.24 and $k_S D = 0.04, 0.2$ and 0.4 . Three
820 frequencies show that an order is not an accident of one.
- 821 7. *The arithmetic.* The errors of the higher-degree voxels fall below the
822 round-off of double precision, so the symbolic implementation computes
823 them to 40 digits; the independent numerical implementation, in double
824 precision, agrees with it down to its own floor of about 10^{-13} .

825 The later sections change one item each: the incident wave to SV or SH and
826 its angle (§8), the medium to a stratified or a graded one and the background
827 to a stratified one (§9), and the body to a sphere (§10).

828 *Second order.* For a layer $D = 2 \text{ m}$ with the contrast $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu =$
829 1 GPa , $\Delta\rho = 100 \text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000 \text{ m s}^{-1}$, $\rho =$
830 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
831 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
832 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (35) and with the
833 package *T-matrix* alike (`ContinuumLimit_Chain.wl`). The error depends on
834 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent

835 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
 836 the exact solution and n the refinement; the orders in the legend of Fig. 7
 837 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
 838 with their standard errors. The two agree where a sequence is asymptotic
 839 and differ where it is not, which the apparent orders show.

840 *The leading error in closed form.* In Eq. (35) the kernel is integrated exactly;
 841 the only approximation is the internal field's variation across a voxel. At first
 842 order in the contrast the discrete scattered field is therefore a midpoint rule
 843 applied to the continuum's Born integral, with the kernel's cell integral exact.
 844 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 845 it is flat. With k the background wavenumber of the wave concerned (k_P for
 846 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 847 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (36)$$

848 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 849 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 850 and those of Eq. (39) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 851 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 852 the transmission constant grows to -0.092 .

853 *Why a uniform-field single site stops at second order.* The two constants have
 854 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 855 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 856 reduce the reflection error by a factor 4.45 and *increase* the transmission

error by 1.7, at every n . The missing term in Eq. (36) is the first moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) = P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (37)$$

with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (38)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form. Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.w1`; Table 6). A single plane of first-moment voxels represents

Table 6: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (38). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

the 2 m layer 6500 times more accurately than a single plane of uniform ones.
The collocation column is the scheme of §6 and reproduces the chain of the
full 9×9 system to all printed digits.

Higher moments. Carrying the Legendre moments to degree p changes nothing in the construction: the moments and the same-cell double integrals of $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-degree solver reproduces the chain above exactly. The second-moment voxel ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at eighth, in reflection and transmission (Table 7; `ContinuumLimit_SecondMoment.w1`); the second-moment voxel’s order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} . The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

The leading error in closed form. Without scattering, the Galerkin solution is the L^2 projection Π of the incident field onto each voxel’s basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel’s cell integral applied to $\Pi w^0 - w^0$. The field

Table 7: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

890 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
891 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
892 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (39)$$

893 with Π the centre value for collocation, the cell mean for the mean-only voxel
894 and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_`
895 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results
896 are in Table 8. The chain of Eq. (38) reproduces the forms for collocation
897 and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being
898 the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7}
899 at a contrast reduced by 10^{-6} . The constant for degree p follows from the
900 Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops
901 the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment
902 against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 8: The leading error of the discrete layer in closed form, Eq. (39): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

903 $h = d/2,$

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (40)$$

904 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
905 mission for odd p , opposite signs for even p . The first-moment voxel's leading
906 error therefore does not depend on the direction in which the field is observed.
907 Eq. (40) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
908 ishes) in Mathematica and, independently, in sympy.

909 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
910 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
911 $d = D$) the series in D of the reflection coefficient, computed exactly at the
912 full contrast, coincides with that of the exact layer, Eq. (7), term by term
913 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
914 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

w1). Each further Legendre moment adds two terms of the thin-layer series.

In three dimensions. The three-dimensional first-moment voxel carries the nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (38) (ContinuumLimit_Reduction3D.w1). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and $x/h, y/h$ are odd there. The mean and the z -moment carry weights that are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift, $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds only to $O(k_{\parallel} h)$.

Other angles, other waves, and degree two. The orders and their constants hold beyond normal incidence and incident P. At oblique incidence, where the tiling identity holds only to $O(k_{\parallel} h)$ and the gradient channels couple through the evanescent orders, the mean-only and first-moment voxels remain second and fourth order in the reflected P and the converted S wave; for incident SV and SH waves the orders, and with $k = k_S$ the constants, are those of incident P; and the voxel of degree two converges at sixth order

940 for P, SV and SH waves, its error on four planes at normal incidence being
941 the $c_2(kd)^6$ of Eq. (40) to 4%. Appendix Appendix B gives the spectral
942 construction, Eq. (B.1), the tests, and the second implementations that check
943 them; Figure 7(a,b) shows the convergence.

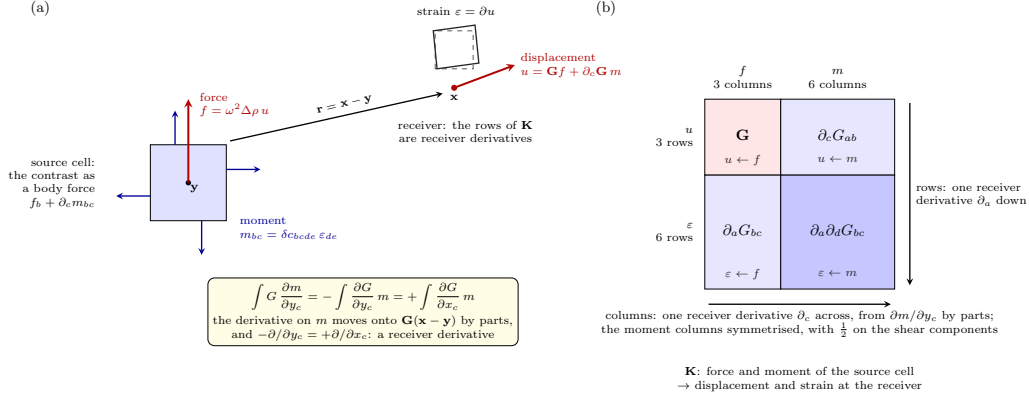


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

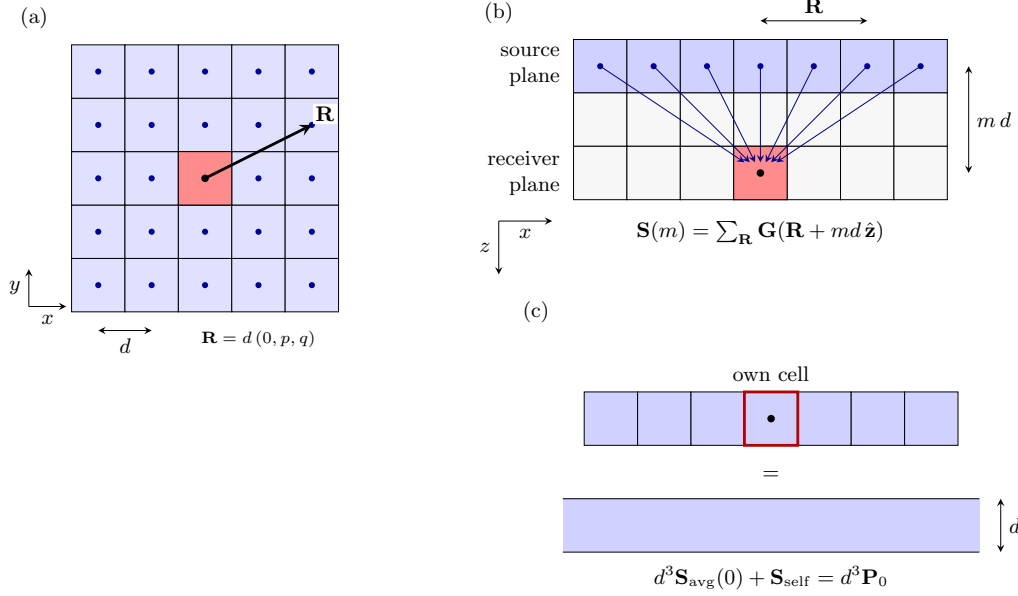


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

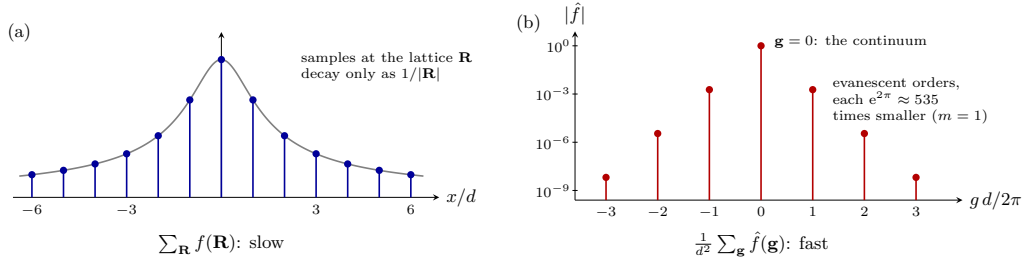


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

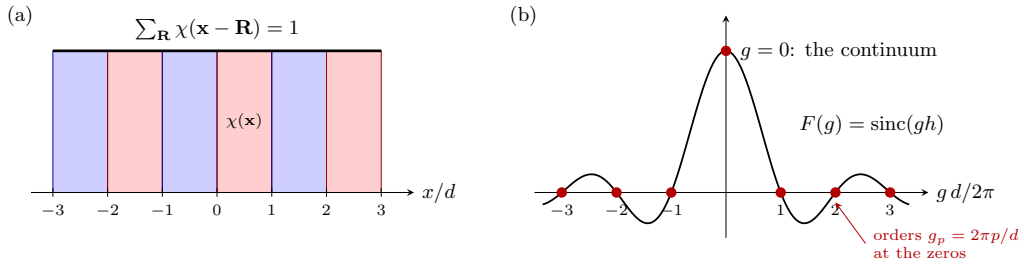


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

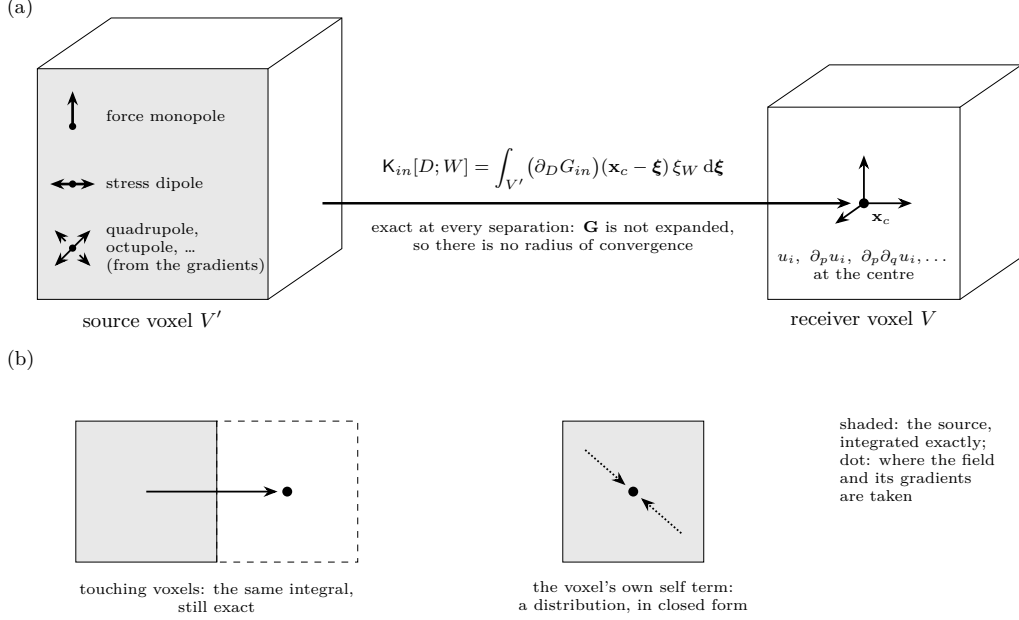


Figure 5: The Cartesian multipole hierarchy. (a) The source voxel radiates through the Cartesian multipole moments of its source: a force monopole from the density contrast, a stress dipole from the stiffness contrast, and the quadrupoles, octupoles and higher multipoles from the field's gradients. The receiver voxel takes the field through its value and gradients at its centre $\mathbf{x}_c$. The two are coupled by the integral of the derivatives of the Green's tensor over the whole source cube, Eq. (47), which is exact at every separation because the Green's tensor is not expanded. (b) The cases a multipole expansion of the Green's tensor cannot reach are therefore included: touching voxels, by the same integral, and the voxel's own self term, a distribution in closed form.

Alt text: Two panels. In the upper panel a shaded cube on the left, labelled source voxel, contains three small symbols: a single arrow labelled force monopole, a pair of opposed arrows labelled stress dipole, and a cross of four arrows labelled quadrupole, octupole, and so on, from the gradients. A long arrow runs from it to the centre of an unshaded cube on the right, labelled receiver voxel, where a dot with three short arrows marks the

field and its gradients at the centre. Above the arrow is the coupling integral of the derivatives of the Green's tensor over the source cube, and below it the words exact at every separation: $\mathbf{G}$ is not expanded, so there is no radius of convergence.

In the lower panel, on the left, a shaded square touches a dashed square whose centre is marked by a dot, with an arrow from the shaded square to the dot, labelled touching voxels: the same integral, still exact; on the right a single shaded square has a dot at its centre and two

944 9. A heterogeneous layer

945 A heterogeneous medium is not a layer of identical cells. The simplest
946 heterogeneous case with an exact answer is a stratified model built from
947 layers of constant properties, with real interfaces between them: the form of
948 many layered crustal velocity models. The last test takes such a model. A
949 layer $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each
950 one $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background
951 (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is
952 a random stratification with physical contrasts. The voxels refine the model's
953 layering, with m planes of voxels in each model layer, so no voxel straddles
954 a model interface: the natural discretisation when the model's interfaces are
955 known. At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so
956 the field reverberates between the model interfaces. The schemes are those
957 of §§7–8, with the contrast now changing from plane to plane; nothing else
958 in them changes. In the terms of the experiment of §7, the test is this.

- 959 1. *The medium.* The eight sets of contrasts are drawn once, from a fixed
960 seed, and the same eight are used in every run, so that every grid and
961 every scheme sees the same model.
- 962 2. *Normal incidence.* The source is the plane force at $z_s = -12$ m, and the
963 scattered displacement is taken at $z = -1$ m, where it is the reflected
964 wave, and at $z = D + 2 = 6$ m, where it is the change in the transmitted
965 wave. The P problem has the modulus contrast $\Delta\lambda + 2\Delta\mu$ and the wave
966 speed α ; the S problem is the same one-dimensional problem with $\Delta\mu$
967 and β .
- 968 3. *Oblique incidence.* A P plane wave is incident at 20° to the vertical.

969 What is measured is the pair of displacement amplitude vectors of
 970 the reflected P and the reflected S plane waves, R_{PP} and R_{PS} , in the
 971 specular order. The lateral sums of the coupling (§8) run over the
 972 reciprocal-lattice orders $|p|, |q| \leq 8$.

973 4. *The exact answer.* The state of displacement and traction, two compo-
 974 nents at normal incidence and four for P–SV at 20° , is carried across
 975 each model layer by the matrix exponential of that layer’s system ma-
 976 trix times its thickness, the eight matrices are multiplied in order,
 977 and the radiation conditions of the two half-spaces are imposed on
 978 the product. The arithmetic carries 30 to 40 digits. The reference
 979 shares nothing with the voxel scheme but the model. It is checked
 980 against a third algorithm, the reflection–transmission recursion of Ken-
 981 nett [67] over the nine interfaces: the same-type coefficients, R_{PP} and
 982 R_{SS} at normal incidence and R_{PP} at 20° , agree to 10^{-14} (`scripts/`
 983 `gate_heterogeneous_reference_vs_kennett.py`). Only same-type
 984 coefficients are compared, because they do not depend on how the up-
 985 and downgoing waves are normalised; a converted coefficient does. One
 986 sign differs, by convention: at normal incidence the one-dimensional
 987 model’s R_{PP} is a ratio of vertical displacements, the recursion’s a ra-
 988 tio of amplitudes along each wave’s own direction, and the reflected P
 989 wave travels upwards.

990 5. *The error.* As in §7, the difference from the exact scattered quan-
 991 tity relative to its size: for the amplitude vectors at 20° , $|\mathbf{R}_{\text{scheme}} -$
 992 $\mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$.

993 6. *The refinement.* Each model layer is cut into m planes of voxels, so

994 that the layer holds $n = 8m$: $m = 1, 2, 4, 8, 16$ at normal incidence
995 and $m = 1, 2, 4$ at 20° , where the lateral sums make each run dearer.
996 The orders are apparent orders, from $m = 4$ to 16 and from $m = 2$ to
997 4.

The errors are in Table 9 and Fig. 7(c).

Table 9: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

998

999 *The interfaces cost nothing.* Every order is the one measured for the uniform
1000 layer. At a model interface the strain of the true field jumps. A voxel basis
1001 that is discontinuous from voxel to voxel represents that jump exactly when
1002 the interface lies on a voxel boundary, so the interface adds no error of its
1003 own. The only requirement is that voxel boundaries coincide with the model's
1004 interfaces. A voxel that straddled an interface would carry a contrast that is
1005 not uniform within it, a case not examined here.

1006 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
1007 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
1008 layer. That figure is not a run: it is the second-order law carried from the
1009 finest grid computed, 128 planes, whose error is 7.3×10^{-6} , to the target,
1010 $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$. The first-moment voxel needs 8, one per model
1011 layer, and that figure is a run: the model's own layering already gives $8.5 \times$
1012 10^{-7} . Counting unknowns (9 per voxel against 36), that is 3123 against 288
1013 per vertical column of voxels, a factor of eleven. In three dimensions, with
1014 the lateral grid refined as the vertical one is, the number of voxels falls by
1015 $(347/8)^3 \approx 8 \times 10^4$ and the number of unknowns by about 2×10^4 . The same
1016 model has been solved independently in Python, with a contrast for each
1017 plane: the raw reflected displacements agree with the Mathematica scheme
1018 to 1.1×10^{-15} at normal incidence and at 20° , with and without the first
1019 moments. Lateral heterogeneity needs the full three-dimensional solver; it is
1020 the first extension discussed below.

1021 *A medium that varies within a cell.* The stratified model is exact on its own
1022 grid: every cell is homogeneous. A smoothly varying medium is not, and
1023 sampling it as cells of constant properties is an approximation of its own.
1024 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
1025 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
1026 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
1027 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (38)

1028 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (41)$$

1029 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 1030 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 1031 the order of convergence is

$$\min(2p+2, 2r+2), \quad (42)$$

1032 or $2p+2$ when degree r represents the profile exactly (Table 10; `ContinuumLimit_`
 1033 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 1034 caps every voxel at second order, however rich its field basis, and for the
 1035 smooth profile the voxel of degree 2 with a linear contrast has the error
 1036 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 1037 this is what one expects: the contrast's projection error is orthogonal to
 1038 the polynomials of degree r , so against the smooth product of the inci-
 1039 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 1040 the contrast must be represented to the degree of the field. The closed forms
 1041 of the construction are tested one by one by an independent implementa-
 1042 tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
 1043 of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
 1044 linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
 1045 double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
 1046 bles the scheme without linearisation, evaluating the projected contrast at
 1047 the quadrature nodes, against its own 30-digit solution of the graded layer;
 1048 its errors agree with those of Table 10 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it

Table 10: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (42), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

1049 measures the orders of Eq. (42) again.

1050 *Which basis sets the error.* Equation (42) gives the order. The size of the
1051 error, and which of the two bases it belongs to, appear when the scattered
1052 field is separated by its order in the contrast: the Born term T_1 , and the
1053 second-order term T_2 in which the field scattered once is scattered again.

1054 Appendix Appendix G gives both terms for the exact layer in closed
1055 form (Eqs. (G.2)–(G.7)) and for the scheme, where the first needs no solve
1056 (Eq. (G.8)), and shows how both are read from solutions at four scaled
1057 contrasts (Eq. (G.9)). The terms must be separated: at the full contrast the
1058 terms of third and higher order are 7% of the second and carry errors of their
1059 own.

1060 With $f(z)$ the profile and Π_q the L^2 projection onto the Legendre poly-
1061 nomials of degree at most q in each cell, define the *relative projection error*

1062 of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (43)$$

1063 the fraction of the squared norm of the profile that polynomials of degree q
 1064 on the cells cannot represent. It is a property of the medium and the grid
 1065 alone; it involves neither the wave nor the solution. For the smooth profile
 1066 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (44)$$

1067 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 1068 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 1069 The error of the Born term is smaller by two to six orders of magnitude. The
 1070 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 1071 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 10
 1072 has 1.4×10^{-7} .

1073 Equation (44) is exact in the long-wave limit (Appendix Appendix G).
 1074 There the strain scattered once is the profile itself, a cell holds it as $\Pi_{\min(p,r)} f$,
 1075 and the second scattering takes place in the cell of the first: the law is
 1076 a statement about the single site, whose T -matrix is exact for the cell as
 1077 represented and whose error is that the cell is not the medium. Measured,
 1078 the part of T_2 between different cells carries at most 0.002 of the error, and
 1079 the departure from the law falls as $(kD)^2$.

1080 Three consequences follow. The two bases must be matched: a contrast
 1081 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 1082 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 1083 the contrast improves only the Born term. The wavefield basis is there to

1084 carry what the medium imprints on the field, not only the wave: at these
 1085 frequencies the wave itself varies little across a cell. And the error is known
 1086 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
 1087 and the cells. It vanishes when the medium is piecewise constant on the cells,
 1088 which is the stratified model above; there the contrast needs no more than a
 1089 constant in each cell, and the degree of the field alone sets the order.

1090 *A stratified background.* Everything above takes the background to be a
 1091 whole space. The method is meant for a background that is plane strati-
 1092 fied, with the voxels carrying the departure from it, and the results must
 1093 hold there. They do, in the one case tested: normal incidence, P, with the
 1094 background two half-spaces joined at $z = z_I$ below the layer, the lower one
 1095 faster and denser (displacement reflection coefficient -0.19). The voxels then
 1096 interact through the Green's function of that background,

$$g(z, z') = \frac{i}{2M_P k} \left[e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (45)$$

1097 with ρ_I the reflection coefficient of the interface: the whole-space term and
 1098 a reverberation that is smooth and carries no local term. The incident field
 1099 is the field of the source in the background, and the exact answer is the
 1100 reflection of the whole stack less that of the background. With the interface
 1101 3 m below the layer, and again with the lowest voxel against it, the scheme
 1102 converges at order 2.00, 4.00 and 6.0 for $p = 0, 1$ and 2 on the uniform
 1103 layer, and at 1.98, 3.98 and 5.98 on the smooth profile with $r = p$. The
 1104 first-order term is predicted to round-off (5×10^{-16}): it is the sum over
 1105 four paths, the incident wave travelling down or up and the scattered wave
 1106 leaving up or down, of the exact first-order term of each path times the

1107 factor $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$ for that path's pair
 1108 of wavenumbers, whose leading terms are the constants of Eq. (40). And the
 1109 relative error of T_2 over $\mathcal{E}_p$ is 1.00 to 1.02 at $kD = 0.12$ on 4 to 16 cells for all
 1110 three degrees, departing from one by 1.1×10^{-4} and 3×10^{-5} at $kD = 0.012$
 1111 for the two positions of the interface: Eq. (44) holds as in the whole space
 1112 (`scripts/measure_layer_stratified_background.py`; the exact stack is
 1113 computed in 30-digit arithmetic). The reason is the one given for Eq. (G.10):
 1114 the reverberation adds no local term, so the second scattering in the cell of
 1115 the first is unchanged. This check is by one implementation against an exact
 1116 solution.

1117 *Oblique incidence, and a three-layer background.* The same holds at oblique
 1118 incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer
 1119 background in which the layer of voxels reverberates between two interfaces
 1120 (Appendix Appendix C). At 20° the orders are 2, 4 and 6 for $p = 0, 1$ and
 1121 2 in all thirty-six cases (Table C.16), and the errors are close to those of the
 1122 whole space: the background changes the field the voxels must represent,
 1123 not how well they represent it. Both terms of the error remain known in
 1124 advance. The scheme's first-order term equals a closed-form prediction, built
 1125 from the thirty-six pairs of incident and radiated plane waves (Eq. (C.3)), to
 1126 2×10^{-15} ; and the law of Eq. (44) holds to 3% for $p \leq 1$ and 6% for $p = 2$ at
 1127 $k_S D = 0.2$, and to 0.6% at $k_S D = 0.067$, in every background and for every
 1128 wave type (Table C.17).

1129 10. A sphere: what the layer predicts in three dimensions

1130 The layer and the stratified layer are laterally uniform. A sphere is not,
 1131 and elastic scattering by a sphere, homogeneous [64, 65] or radially varying
 1132 [68], has an exact solution: it is the usual benchmark for volume-integral and
 1133 discrete-dipole methods. This section solves two spheres with the full three-
 1134 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
 1135 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
 1136 iterative solver, and a P wave is incident. For the graded sphere the voxel is
 1137 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
 1138 nine-component state, tested with the same constant, and the L^2 projection
 1139 of its contrast, which is its cell mean; every cell that overlaps the sphere is
 1140 kept, and the cells are coupled by cell-to-cell double integrals of the Green’s
 1141 tensor, one block per orbit of the cube group (Appendix Appendix E). For
 1142 the sharp sphere the voxel is the package’s cube T -matrix, which differs from
 1143 the collocation closure only through its far-field corrections (§6.6) and has
 1144 the same order, each cell keeping the contrast of its centre. The far field is
 1145 computed from the solution for the incident plane wave itself, each voxel’s
 1146 displacement and strain carrying the plane-wave phase at its centre; adding
 1147 to that phase a linear variation about the scatterer’s centre, as a Taylor
 1148 expansion of the incident field would, counts the wave’s variation twice and
 1149 leaves an error of order $(ka)^2$ that no refinement removes. In the terms of
 1150 the experiment of §7, the test is this.

- 1151 1. *The body.* A sphere of radius $a = 10$ m (Fig. 6) in the background of
 1152 §7, carrying the contrast of that section, at $\omega = 150$ and 300 rad s^{-1} ,
 1153 for which $k_S a = 0.5$ and 1 and $k_P a = 0.3$ and 0.6 .

- 1154 2. *The incident wave.* A P plane wave travelling along $+z$.
- 1155 3. *The grid.* The cube of side $2a$ that circumscribes the sphere is cut into
1156 n_{sub}^3 cells of side $2a/n_{\text{sub}}$; which of them are kept is stated with each
1157 voxel above.
- 1158 4. *The solve.* The iteration is carried to a relative residual of 10^{-11} , far
1159 below every discretisation error reported, so that the errors are the
1160 scheme's and not the solver's.
- 1161 5. *The exact answer.* The partial-wave series of the sphere, summed to
1162 order 10 at $k_S a = 0.5$ and 11 at $k_S a = 1$, and evaluated as the exact
1163 scattered displacement at the observers, not as its asymptotic form.
- 1164 6. *What is measured.* The scattered displacement vector at nine scattering
1165 angles, equally spaced from 11° to 169° in the z - x plane: forward and
1166 backward scattering and everything between, P and S together.
- 1167 7. *The error.* The largest component of the difference between the scheme's
1168 far field and the exact displacement over the nine angles, divided by the
1169 largest component of the exact one. It is a maximum, not an average,
1170 so that no angle is hidden.
- 1171 8. *The distance.* The solution of the scheme is the field inside the voxels,
1172 and the scattered field outside follows from it at any distance. For the
1173 far-field test it is read with the asymptotic formula, which keeps the
1174 term in $1/r$ alone: a choice of this test, not a property of the scheme.
1175 The reference is the exact field, so the two differ by the near-field terms,
1176 of relative size $1/(kr)$, however good the scheme. The graded sphere
1177 is therefore observed at 5×10^8 radii, where those terms are 10^{-8} of
1178 the peak: at 5×10^4 radii they are 1.4×10^{-4} , which is 4% of the

1179 discretisation error at 12 voxels across and most of it at 24, and would
 1180 be read as a failure to converge. The sharp sphere, whose errors are
 1181 a hundred times larger, is observed at 5×10^4 radii. The near field is
 1182 tested separately, with a readout that drops nothing (Table D.18).

1183 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 1184 its shape with every grid. A sphere whose contrast falls smoothly to zero at
 1185 its surface does not: the voxels that cross the surface carry a vanishing con-
 1186 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 1187 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 1188 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 1189 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 1190 the radial equations of Takeuchi and Saito [68], integrated across the shell
 1191 from the core’s regular solutions and matched to outgoing waves at $r = a$.
 1192 The equations are derived symbolically from the equations of motion, and
 1193 the solution is checked against a sphere of homogeneous concentric shells
 1194 in closed form, by unitarity and reciprocity, and by an independent imple-
 1195 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 1196 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 1197 carries the mean of the contrast over its cell. Table D.19 shows the error
 1198 falling at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/`
 1199 `pilot_graded_voxel_sphere.py`): the apparent order between successive
 1200 grids is 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares
 1201 orders of Fig. 7(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across
 1202 the error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 1203 graded shell and the order is not yet asymptotic. This is the order the layer

1204 predicts for the mean-only voxel, and the first test of that prediction on a
1205 body with genuine three-dimensional structure.

1206 What makes the sequence this clean is that the voxel carries a single error.
1207 The collocation voxel of the sharp sphere applied to this body, each cell at
1208 the profile's value at its centre and only the cells whose centres lie inside the
1209 sphere kept, carries two: its own internal-field error and the error of sampling
1210 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
1211 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
1212 2.0, 3.5 and 3.6 at $k_{\text{sa}} = 0.5$, and at $k_{\text{sa}} = 1$ the apparent orders are 1.7, 2.5
1213 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
1214 but its apparent order is not that of any single term and cannot be read from
1215 them. The same collocation voxel with the cell mean of the contrast, and
1216 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
1217 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
1218 second term, and with it removed the collocation and mean-only voxels differ
1219 only by their constants, as in Table 8. The cell mean, with every overlapping
1220 cell kept, represents the same body on every grid, and the order the layer
1221 predicts is then the one observed.

1222 The first-moment voxel of §8, extended to three dimensions with a con-
1223 trast linear in each cell, converges at fourth order on this same sphere (Ap-
1224 pendix Appendix E): its apparent orders are 4.01 and 4.13 where the dis-
1225 cretisation error is isolated at weak contrast, and at the full contrast its error
1226 with 16 voxels across is 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below
1227 the mean-only voxel's on the same grid.

1228 *The near field, an impedance march, and a sharp sphere.* Three further tests
 1229 are in Appendix Appendix D. Read at finite distance, down to a quarter
 1230 of a radius from the surface, the graded sphere converges at the orders of
 1231 the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-
 1232 moment voxel, while the asymptotic readout would be in error there by 48%
 1233 to 85% (Table D.18). Against the impedance march of Haines et al. [69],
 1234 which needs far fewer unknowns, the first-moment voxel reaches the march's
 1235 accuracy with six voxels across and continues at close to fourth order below
 1236 it, where the march is limited by the scattering between the spheres of the
 1237 periodic array it must solve (Table D.20). And a sphere with a sharp surface
 1238 does not converge: its error tracks the volume of its staircase, which changes
 1239 with the grid (Table D.21); the smooth sphere, like the layer, measures the
 1240 scheme.

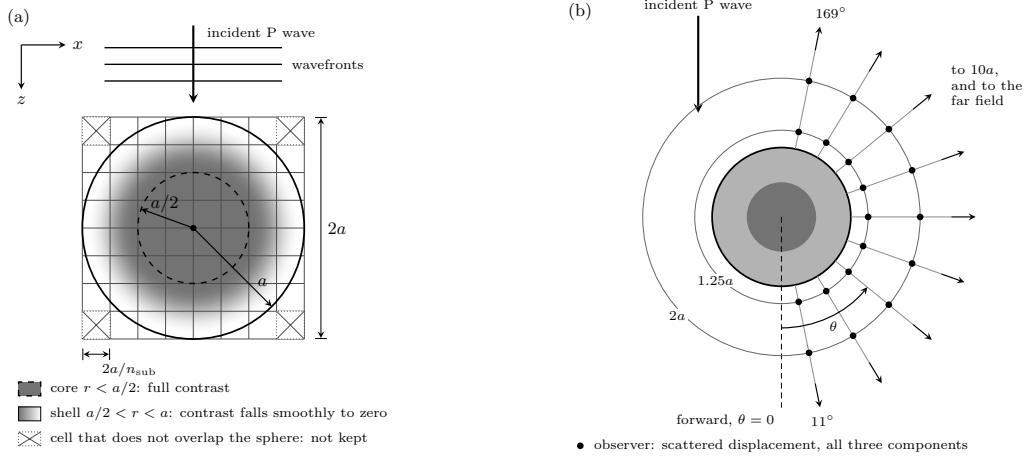


Figure 6: The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core $r < a/2$ and falls smoothly to zero across the shell $a/2 < r < a$. The circumscribing cube of side $2a$ is cut into n_{sub}^3 cells, here $n_{\text{sub}} = 8$; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along $+z$. (b) The observers. The scattered displacement is taken at nine scattering angles θ , measured from the forward direction and equally spaced from 11° to 169° , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

Alt text: Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

11. The Cartesian multipole hierarchy as a voxel scheme

The Legendre voxels of §8 represent the field of a voxel by its moments over the voxel. The hierarchy of §6 represents it by its gradients at the centre. Both are polynomial, and §6.4 shows that on the layer the hierarchy of degree $\ell = 3$ is fourth order, as the first-moment voxel is. This section builds the hierarchy into a voxel scheme in three dimensions and compares the two.

The scheme. Each voxel c , centred at $\mathbf{x}_c$, carries the coefficients $U_{j;W}^c$ of Eq. (20). The equation and its derivatives to order ℓ are imposed at every centre (with $\ell = 1$ and no density contrast this is the static numerical equivalent inclusion method on cuboidal elements, with uniform eigenstrains, equivalence at the element centres and a fast Fourier transform, of Zhou et al. [70]):

$$U_{i;P}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_P u_i^0(\mathbf{x}_c), \quad (46)$$

$$K_{in}[D; W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (47)$$

The two kinds of block are of different character. For $c' = c$ the integrand is singular and the integral is a distribution: it is the moment of the hierarchy, $K_{in}[D; W](0) = (-1)^{|D|} \mathbf{g}_{in}[D; W]$, in closed form on the cube. For $c' \neq c$ the field point is the centre of another voxel, at least half a voxel from the source voxel, so the integrand is smooth and a tensor Gauss rule integrates it; the derivatives of the Green's tensor, to any order, come from the family $x^a y^b z^c r^p e^{ikr}$, which is closed under differentiation. All the singular work of the scheme is thus in the self block, where it is done once and in closed

1263 form; the Legendre voxel's self block is a double integral of the same singular
 1264 kernel. The blocks depend on the offset alone, so the product of the system
 1265 with a vector is a convolution over the lattice and is done exactly by the fast
 1266 Fourier transform. Where the contrast varies within a voxel it is projected
 1267 there on polynomials of degree r , as in §9, and the source is the product of
 1268 that polynomial and the field's: the moments then carry weights to degree
 1269 $\ell + r$, and a block is a sum, over the monomials of the contrast, of blocks
 1270 with the weight shifted.

1271 *Checks of the construction.* The derivatives of the Green's tensor to fourth
 1272 order agree with the closed forms of its value, gradient and second derivative,
 1273 and with central differences beyond them, to 8×10^{-7} . A lattice of one voxel
 1274 reproduces the single site of §6.5 to 10^{-14} . Only the offsets of one octant
 1275 are computed, the others following from the parity of each index; a block
 1276 obtained that way equals one computed directly to 10^{-15} . The iterative
 1277 solution equals a dense solution to 7×10^{-11} where both can be run.

1278 *A body with corners.* The cube of §6.5, at $k_{sh} = 0.2$, is cut into n^3 voxels.
 1279 Table 11 gives the far-field error against the converged reference, with the
 1280 Legendre voxels on the same body. On this body every degree of the hierarchy
 1281 converges at about the same rate, an order near two between the two finest
 1282 grids, and the degree buys a constant: thirty times from $\ell = 1$ to $\ell = 3$
 1283 on 64 voxels. That rate is the body's and not the scheme's, for the mean-
 1284 only Legendre voxel shows it too: a homogeneous cube has edges and corners,
 1285 whose fields no polynomial follows. The uniform-strain closure and the mean-
 1286 only voxel agree on one cell, as they must, being the same approximation.
 1287 And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is

Table 11: A homogeneous cube cut into n^3 voxels, $k_S h = 0.2$: relative far-field error of the hierarchy of degree ℓ and of the Legendre voxels, with the unknowns of each voxel.

	unknowns a voxel	$n = 1$	2	3	4
hierarchy, $\ell = 1$	12	5.4×10^{-3}	1.7×10^{-3}	8.3×10^{-4}	5.0×10^{-4}
hierarchy, $\ell = 2$	30	1.6×10^{-3}	6.7×10^{-4}	3.7×10^{-4}	2.4×10^{-4}
hierarchy, $\ell = 3$	60	3.4×10^{-4}	7.0×10^{-5}	3.0×10^{-5}	1.7×10^{-5}
Legendre, mean only	9	5.4×10^{-3}	1.3×10^{-3}	5.8×10^{-4}	3.3×10^{-4}
Legendre, first moment	36	4.8×10^{-5}	6.0×10^{-6}		

more accurate than eight voxels of the hierarchy of degree three with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [54] established for the static equivalent inclusion method.

A smooth medium. The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §6.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile $\sin^2(\pi z/D)$ the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for $\ell = 1, 2$ and 3: a varying contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. The graded sphere of §10 has a shell

1304 5 m wide, and a solver that reaches fourteen voxels across puts fewer than
1305 four voxels across it; so the core is reduced here from $a/2$ to $a/10$, which
1306 widens the shell to 9 m, and a second profile, $\sin^2(\pi x/2)$ in the fraction x of
1307 the shell's width measured from the surface, is run beside the smoothstep.
1308 The exact reference is the radial solution of §10 for each body. The contrast
1309 is linear in each voxel for $\ell = 3$, which suffices for fourth order (§9), and of
degree ℓ for $\ell = 1$ and 2.

Table 12: The graded sphere with a shell 9 m wide, $k_{Sa} = 0.5$: relative far-field error
against the exact solution for n_{sub} voxels across, with the apparent order between the two
finest grids of each row. The lower block is the profile $\sin^2$.

	$n_{\text{sub}} = 4$	6	8	10	12	14	order
hierarchy, $\ell = 1$	1.5×10^{-3}	8.4×10^{-4}	4.9×10^{-4}	3.2×10^{-4}			1.94
hierarchy, $\ell = 2$	3.3×10^{-4}	2.1×10^{-4}	1.5×10^{-4}	1.0×10^{-4}			1.56
hierarchy, $\ell = 3$	2.6×10^{-4}	5.0×10^{-5}	1.7×10^{-5}	7.2×10^{-6}	3.6×10^{-6}	2.0×10^{-6}	3.80
Legendre, mean only	7.3×10^{-3}	3.2×10^{-3}	1.8×10^{-3}	1.1×10^{-3}			2.02
Legendre, first moment	2.9×10^{-4}	6.4×10^{-5}	2.2×10^{-5}	9.4×10^{-6}	4.6×10^{-6}	2.5×10^{-6}	3.92
hierarchy, $\ell = 3$	1.9×10^{-4}	3.8×10^{-5}	1.3×10^{-5}	5.2×10^{-6}	2.7×10^{-6}	1.5×10^{-6}	3.84
Legendre, first moment	2.2×10^{-4}	4.8×10^{-5}	1.6×10^{-5}	6.8×10^{-6}	3.4×10^{-6}	1.8×10^{-6}	4.02

1310

1311 Table 12 gives the result. The first-gradient scheme converges at second
1312 order, and is three to four times more accurate than the mean-only Legendre
1313 voxel on every grid, for twelve unknowns a voxel against nine. The second-
1314 gradient scheme is more accurate again but gains no order, as §6.4 predicts;
1315 its apparent order is still rising towards two. The third-gradient scheme
1316 converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8

1317 over the range from six to fourteen voxels across, for both profiles; the first-
 1318 moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over
 1319 the range. Neither has settled on four at fourteen voxels across, and the
 1320 second profile, gentler than the first, changes neither: the two fourth-order
 1321 schemes behave alike. On every grid from six voxels across the hierarchy's
 1322 error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel
 1323 against thirty-six, so that for a given number of unknowns the Legendre voxel
 1324 remains somewhat the more accurate; the wide margin of Table 11 was the
 1325 corners.

1326 *What the hierarchy is for.* The two schemes reach the same order by different
 1327 closures of the same equation, and they differ in what they cost and in what
 1328 they give besides. The Legendre voxel is the more economical in unknowns.
 1329 The hierarchy's blocks are simpler: its self block is known in closed form on
 1330 four constants (§6.2), and its coupling is a single integral of a smooth func-
 1331 tion where the Legendre voxel's is a double integral. The iteration converged
 1332 in five steps at every size; the time goes into the tables of blocks, which are
 1333 computed once for a grid and a frequency. Formed on the same footing, one
 1334 block for each orbit of offsets under the 48 symmetries of the cube, the others
 1335 following exactly, and every point value of the Green's tensor or its derivatives
 1336 from the same compiled radial ladder, the hierarchy's tables are the cheaper:
 1337 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and 23.3 s for
 1338 the first-moment Legendre voxel, with eight times fewer kernel evaluations.
 1339 Both have a one-off cost besides, the hierarchy's the smaller (0.3 s against
 1340 2.2 s): the moments of its self block depend on neither the frequency nor the
 1341 cell size, and are computed once and stored (`scripts/build_cube_unit_`

1342 `moments.py`; `scripts/measure_hierarchy_table_cost.py`; one machine,
1343 five threads). And the hierarchy is, in addition, a single-site T -matrix in
1344 closed form, whose accuracy improves block by block (§6.5), which no voxel
1345 defined by moments over the cell provides. The lattice results of this section
1346 come from one implementation, checked against exact solutions and, where
1347 it can be, against a dense solve; they are at one frequency and on one fam-
1348 ily of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_`
1349 `lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_`
1350 `hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.`
1351 `py`).

1352 12. Discussion

1353 Figure 7 collects the convergence of every test, with the order fitted to
1354 each by least squares (`scripts/plot_convergence_orders.py`).

1355 *What each basis is for.* A voxel holds two functions, and the results give
1356 each basis its part. The basis of the wavefield sets how well the voxel carries
1357 the field across itself: in a uniform medium the scheme converges at order
1358 $2p + 2$ with the constant c_p of Eq. (40), for P and S waves and at normal and
1359 oblique incidence alike. The basis of the medium sets how well the voxel *is*
1360 the medium: with degree r the order is capped at $2r + 2$ (Eq. (42)). Neither
1361 can make up for the other, and the second-order term shows why. The field
1362 scattered once is a copy of the medium’s profile, so the wavefield basis must
1363 carry the profile as well as the wave, and the error of that term is the relative
1364 projection error of the profile at the lower of the two degrees (Eq. (44)). At
1365 the frequencies of interest the wave varies little across a cell and the medium

1366 may vary much, so it is the medium, not the wavelength, that decides the
 1367 degree and the size of the cells. Two cases bracket the choice. Where the
 1368 medium is piecewise constant on the cells, as in a layered model with voxel
 1369 faces on its interfaces, a constant is exact for the medium, the projection
 1370 error vanishes, and the degree of the field is spent entirely on the wave.
 1371 Where the medium varies smoothly within the cells, the two degrees should
 1372 be equal: a higher degree for the contrast than for the field is wasted, and a
 1373 higher degree for the field improves only the Born term. That the two degrees
 1374 can be chosen independently is the practice of higher-order electromagnetic
 1375 volume methods [1]; what the present results add is the rule for choosing
 1376 them, and the error each choice leaves.

1377 *Where the error of a voxel scheme lives.* The results separate the two ap-
 1378 proximations a voxel scheme makes. The *coupling* between voxels makes
 1379 none: averaged over the source cell it is the continuum's cell integral exactly,
 1380 between planes by the form factor's zeros and within a plane by the tiling
 1381 identity, and the self term it omits is exactly the one the single site carries.
 1382 Every error therefore comes from the *single site*: from the internal field it
 1383 assumes and, where the medium varies within the cell, from the medium it
 1384 is given. Its T -matrix sums the rescattering within the voxel exactly, but
 1385 for the voxel as represented, and the split of the second-order term in §9
 1386 measures the consequence: the error sits in the part whose two scatterings
 1387 lie in one cell, and the part between cells carries at most 0.2% of it. For a
 1388 uniform field that error is the midpoint rule's, known in closed form; for a
 1389 linear field it is the next term of the same expansion. The converse is the
 1390 practical warning. A kernel that is not the cell integral (the point Green's

1391 tensor, or a cell average that does not match the single site’s closure) leaves
 1392 a local term that does not vanish as the voxels shrink (§4): the evanescent
 1393 lattice orders carry a static contribution of fixed size per voxel volume. A
 1394 scheme built that way converges to the wrong limit, and refinement cannot
 1395 reveal it, because the error is scale-invariant.

1396 *What the single site is.* The Cartesian multipole hierarchy of §6 turns the
 1397 uniform-field single site from a modelling choice into a derived object with
 1398 a stated error (§6.7). Its one departure from exact scattering by a sphere,
 1399 the real $(ka)^2$ term of Eq. (F.1), and the collocation error of Eq. (36) have
 1400 the same origin, the variation of the field across the scatterer, and the same
 1401 remedy, a richer internal field. In a lattice the remedy is cheap: the field
 1402 a voxel must represent is the smooth continuum field, not the field of an
 1403 isolated cube, with its concentrations at the edges and corners.

1404 *Relation to other methods.* The partition of the error is the coupled-dipole
 1405 method’s choice of polarisability seen from the other side: for space-filling
 1406 cubes the tiling identity fixes the local field exactly, and the polarisability
 1407 is neither needed nor a free parameter. The voxel of degree p is close kin
 1408 to the spectral element of seismology; the two share a Galerkin condition
 1409 on polynomial bases, and differ as a differential formulation from an integral
 1410 one, and so suit different problems. Appendix Appendix H sets out both
 1411 relations, and where each method is the better choice.

1412 *The tension between the two error sources.* The coupled-dipole method has
 1413 two sources of error, the single site and the coupling between sites, and each
 1414 has been corrected on its own: the polarisability by the Clausius–Mossotti

1415 relation, the lattice dispersion relation or the exact static limit, and the
 1416 coupling by integrating or filtering the Green tensor. The two are not in-
 1417 dependent. The point kernel's evanescent orders carry a static local term
 1418 whose size per voxel volume does not shrink with the voxels (§4), and the
 1419 single site must cancel exactly that term. A polarisability tuned for one
 1420 kernel is therefore wrong for another: the lattice dispersion relation corrects
 1421 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 1422 tered one changes the term it must cancel. Tuning either alone moves the
 1423 error between the two, and refinement cannot reveal a mismatch, because
 1424 the surviving term is scale-invariant. For space-filling cubes the tension is
 1425 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 1426 term that the closure carries (Eq. (35)), so the coupling error vanishes iden-
 1427 tically and every remaining error is the single site's, the variation of the field
 1428 across the cell. That error has a second tension of its own. It is a midpoint-
 1429 rule error whose constants have opposite signs in reflection and transmission
 1430 (Eq. (36)), so any scalar adjustment of the single site that improves one wors-
 1431 ens the other; the package's far-field corrections reduce the reflection error
 1432 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 1433 however chosen, can remove it. What removes it is a richer internal field:
 1434 the first moment lowers the error by two orders in kd in both observables at
 1435 once.

1436 *The cost of higher order, and a fast solver.* A first-moment voxel carries four
 1437 times as many unknowns as a uniform one, but by the closed-form constants it
 1438 reaches a reflection error of 10^{-6} with voxels 58 times larger in each direction;
 1439 the stratified model of §9 measures it directly, 347 planes of uniform-field

1440 voxels against 8 first-moment voxels. And because the discretisation error
 1441 is known in advance and the coupling between cells is exact, a fast product
 1442 need be no more accurate than that error, and can be budgeted separately
 1443 from it. Appendix Appendix I sets out the cost, and what the error analysis
 1444 offers a fast solver.

1445 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 1446 mogeneous layer and a randomly stratified one, for which the discrete system
 1447 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 1448 moment voxel in a full three-dimensional solver, where the gradient channels
 1449 couple through the evanescent lattice orders (§8) and the tiling identity holds
 1450 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 1451 voxel converging at second order on a laterally heterogeneous body against
 1452 an exact solution, and the first-moment voxel converges at fourth order there
 1453 (§11). A contrast that varies within a cell, which §9 shows is needed beyond
 1454 second order in a smoothly varying medium, has a further consequence in
 1455 three dimensions: its linear part is odd about the cell’s centre, so it couples
 1456 the single site’s uniform-field and first-moment blocks, which parity keeps
 1457 apart for a homogeneous cube (§6.3). A graded voxel therefore needs at least
 1458 the first-moment block; that voxel is built, and tested on the graded sphere,
 1459 in the companion paper. The numerical evidence is at moderate, physically
 1460 typical contrasts (up to 15% in density and moduli) and at $k_S h \leq 0.3$, the
 1461 range in which the single-site T -matrix is validated; the closed forms are first
 1462 order in the contrast, and the departures from them at the full contrast are of
 1463 second order. What is established is the *order* of convergence and its leading
 1464 constant, not the behaviour at strong contrast, where the single site’s own

1465 closure may limit the accuracy before the lattice coupling does. The single
 1466 site's dynamic error is in closed form for a sphere only; for a cube, for which
 1467 no exact solution exists, it is known through the layer (§§7–8) and, for the
 1468 isolated cube, by comparison with a converged solution (§6.5). The hier-
 1469 archy beyond the first gradient is established less widely than the rest: its
 1470 orders and its error factor on the layer are derived symbolically and checked
 1471 independently to the third degree, and its moments are confirmed by two
 1472 routes, but its three-dimensional results, the single site against the sphere
 1473 and the cube and the voxel scheme of §11, come from one implementation, at
 1474 one frequency for the voxel scheme, checked against exact solutions. There
 1475 the fourth order is approached and not attained on the grids reached (3.8
 1476 at fourteen voxels across, as for the first-moment voxel), the fifth-gradient
 1477 system, whose moments are in hand, has not been assembled in three di-
 1478 mensions, and the two fourth-order schemes have been compared in the cost
 1479 of forming their tables only (§11). Two extensions follow directly. The first
 1480 is the three-dimensional solver itself. The second is a stratified *background*,
 1481 a layered reference medium, as distinct from the stratified contrast of §9.
 1482 For a laterally uniform layer of voxels the orders, the first-order factor and
 1483 the projection-error law all survive it, at normal and at oblique incidence,
 1484 for P, SV and SH waves, in a background of two half-spaces and in one of
 1485 three layers (§9). What remains is a body of finite lateral extent in such a
 1486 background, where the image of a voxel in an interface it touches is singular.

13. Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree p and the contrast to degree r in each voxel, the scheme converges at order $\min(2p+2, 2r+2)$, with the leading error $c_p(kd)^{2p+2}$ in closed form (Eq. (40)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree $\min(p, r)$: it is the single site's, whose T -matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a Cartesian multipole hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [53] to bodies that are not ellipsoids, whose

1512 moments are in closed form on four constants. Its blocks gain accuracy
1513 in pairs, with the error of each truncation in closed form: as a single site
1514 the third-gradient system is three hundred times more accurate than the
1515 uniform-strain closure against the exact sphere, and as a voxel scheme it
1516 is fourth order, as the first-moment voxel is, on a smoothly graded sphere,
1517 where in a lattice the Legendre voxel is the more economical. The results
1518 and the evidence for each are collected in Appendix Appendix J.

1519 *The error is known in advance.* For every discretisation studied here the
1520 error is not something to be discovered by refining until the answer stops
1521 changing. Its order and its leading constant are known before anything is
1522 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
1523 collocation voxel (Eq. (36)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
1524 voxels of degree 0, 1 and 2 (Eq. (40)), and the order $\min(2p + 2, 2r + 2)$
1525 where the medium varies within a cell (Eq. (42)), with the size of the error
1526 there given by the relative projection error of the medium (Eq. (44)), which
1527 is computed from the model and the grid alone. That is worth more than
1528 the order alone, for three reasons. First, the cell size and the degree of the
1529 voxel that a required accuracy needs follow from the wavenumber and the
1530 medium, before the computation and without a refinement study. Secondly,
1531 it turns a refinement study from a hope into a test. A scheme of this kind can
1532 converge smoothly to the wrong limit (§4), and smooth convergence alone
1533 cannot reveal it; but the difference between two grids must have the known
1534 order and the known constant, and a computation whose differences do not
1535 is in error from some cause other than the discretisation. Thirdly, it shows
1536 where effort pays. The error is that of representing the field and the medium

1537 within a cell, so raising the degree of the cell does what multiplying the
 1538 cells cannot: in the stratified model one first-moment voxel per layer reaches
 1539 10^{-6} where the uniform-field voxel needs about forty. Whether the error of
 1540 a body with three-dimensional structure can be known in advance in the
 1541 same way is the question taken up in a following paper [71], where it decides
 1542 where the cells of an adaptive grid are placed. That paper continues the
 1543 paper's first strand, the two bases and their error, with the Legendre voxels.
 1544 The second strand, the Cartesian multipole hierarchy, whose coefficients are
 1545 integrals of the Green's tensor over the cube, is taken further in a paper on
 1546 the coupling integrals of polynomial cubes in closed form, of which it is the
 1547 worked application.

1548 **Appendix A. The moment engine: gates and a second route**

1549 Two checks, whose answers do not depend on the engine, gate every grade.
 1550 Because $\nabla^2 r^m = m(m+1)r^{m-2}$ and $\nabla^2(1/r) = -4\pi\delta$,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi (-1)^{|R|} (\partial_R x_W)(0),$$

(A.1)

1551 for any further derivatives R and any weight W . The second is the sum rule
 1552 (27) at every grade: it tests the δ function of the inclusion problem and its
 1553 derivatives, which is where a quadrature fails. Both hold for every grade of
 1554 Table 2 with two or more derivatives, 910 traces of each kind; the grades with
 1555 fewer, for which the integrand is integrable, agree with numerical integration
 1556 over the cube.

1557 The closed forms are also confirmed by a route that shares nothing with
 1558 the faces of the cube. The cube is its inscribed ball plus the remainder.

1559 On the ball the derivatives are moved onto the sphere, where the surface
 1560 integrals of monomials are Gamma functions, and all the distributional con-
 1561 tent is there; on the remainder the integrand is an ordinary function, in-
 1562 tegrated radially in closed form and over directions through the six faces
 1563 by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored
 1564 values, the largest difference being 1.1×10^{-12} , and satisfies Eq. (A.1) by
 1565 itself to 5×10^{-10} or better up to eight derivatives (`CubeMomentCore.wl`,
 1566 `CubeMomentHigherGrades.wl`, `CubeMomentStore.wl`, `CubeMomentCanonical.`
 1567 `wl`; `scripts/crosscheck_cube_moments_ball_shell.py`).

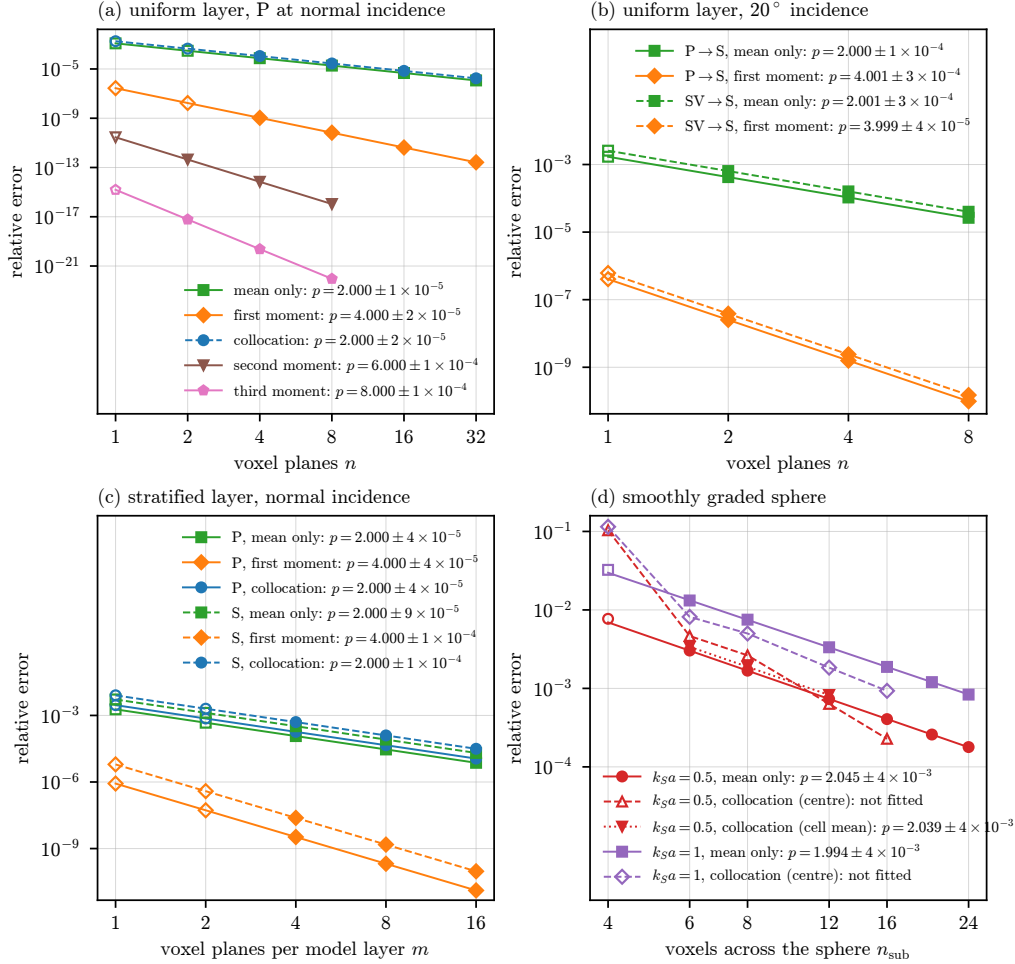


Figure 7: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (38) and (B.1)). (a) The uniform layer, P reflection at normal incidence ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 8; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and

1568 Appendix B. Further tests of the Legendre voxel

1569 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
1570 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (\text{B.1})$$

1571 with L_a the lateral Legendre factors. The whole-space tensor decays as
1572 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
1573 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma +$
1574 $\mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
1575 averaging makes the lattice sum converge without Ewald splitting, to three
1576 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
and 40° (`ContinuumLimit_Oblique.wl`; Table B.13). The mean-only voxel

Table B.13: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (B.1). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

1577
1578 is second order and the first-moment voxel fourth order in both the reflected
1579 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$

1580 and the evanescent coupling of the gradient channels do not cost an order.

1581 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (38) to all printed digits.

1582 *Incident S.* Nothing in the construction is specific to P. For a normally in-
 1583 cident S wave the layer is the one-dimensional problem of Eq. (38) with
 1584 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
 1585 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
 1586 transmission, and the full 9×9 construction with an incident SH wave re-
 1587 produces that one-dimensional result to all printed digits. At oblique inci-
 1588 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
 are those of Table B.14. The orders, and with $k = k_S$ the closed-form con-

Table B.14: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

1589

1590 stants of Eqs. (36) and (39), are those of the incident P wave.

1591 *Degree two at oblique incidence.* In three dimensions the voxel of degree
 1592 2 carries, on each of the nine components of its state, the ten Legendre
 1593 products of total degree at most two in (z, x, y) . In the construction of
 1594 Eq. (B.1) it converges at sixth order against the exact layer for incident
 1595 P, SV and SH waves (Table B.15; `scripts/measure_oblique_degree.py`).

1596 The test is run at $\omega = 3000 \text{ rad s}^{-1}$ ($k_P D = 1.2$, $k_S D = 2$), ten times the
1597 frequency of the other tables, because at 300 rad s^{-1} a sixth-order error on
1598 these grids lies below the round-off of double precision. The exact reflection
1599 coefficient is that of Kennett's recursion, an independent algorithm, and the
1600 same computation with the bases of degree 0 and 1 returns orders 2 and 4
1601 in all five cases. At normal incidence the error on four planes, 6.95×10^{-9} , is
1602 the $c_2(kd)^6 = 7.2 \times 10^{-9}$ of Eq. (40) to 4%. Doubling the number of lattice
1603 orders retained in Eq. (B.1) changes the errors by less than 1%. Unlike the
1604 degrees 0 and 1, this degree is implemented once at oblique incidence; its
check is the exact solution.

Table B.15: The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000 \text{ rad s}^{-1}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

incident wave	error, $p = 2$			order	orders
	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0 / p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20° : R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20° : R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20° : R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40° : R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

1605

1606 The spectral construction is implemented twice. The second implementa-
1607 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
1608 nition of the scheme: it takes the depth dependence of the transformed kernel
1609 from an independent residue construction, obtains the local weight as a nu-

1610 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
 1611 ble integral by quadrature and separates P from S by sampling the upgoing
 1612 field at two depths. The raw reflected displacements of the two agree to $9 \times$
 1613 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
 1614 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
 1615 scheme of any degree is also implemented twice. The second implementation
 1616 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
 1617 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
 1618 from its own interface system, in double precision. Its errors agree with the
 1619 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 1620 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 1621 6.00; for one plane as the layer thins its differences from the exact layer fall
 1622 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 1623 Figure 7(a,b) shows the convergence, with the orders fitted to it.

1624 **Appendix C. The stratified-background construction**

1625 The experiment is built in four steps.

1626 *The kernel.* At each lateral wavenumber $\boldsymbol{\kappa}$ the whole-space kernel between
 1627 two depths factorises over the plane-wave modes, three going down and three
 1628 going up (P, SV, SH): $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') = \mathbf{D}_\downarrow \mathbf{E}(z - z') \mathbf{S}_\downarrow$ for $z > z'$, with $\mathbf{S}_\downarrow$ the
 1629 3×9 matrix of mode amplitudes radiated downwards by a nine-component
 1630 source, $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$ the propagation over a distance L with γ_m the
 1631 vertical wavenumber of mode m , and $\mathbf{D}_\downarrow$ the 9×3 matrix of the modes'
 1632 states; and likewise upwards. With the interfaces at z_A above and z_B below,
 1633 $H = z_B - z_A$, and $\mathbf{R}_A, \mathbf{R}_B$ the 3×3 reflection matrices of the two interfaces

1634 seen from inside the layer, a source at z' sends $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow q$ down and $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow q$
 1635 up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\boldsymbol{\kappa}; z, z') q &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \tag{C.1}$$

1636 with $\mathbf{d}_A$ the downgoing amplitudes leaving the upper interface and $\mathbf{u}_B$ the
 1637 upgoing ones leaving the lower. It is added to the whole-space kernel in
 1638 every block of Eq. (B.1), at every order of the reciprocal lattice. It is smooth
 1639 and separable in z and z' , so its integrals over the cells are one-dimensional
 1640 moments of exponentials. For two half-spaces $\mathbf{R}_A = 0$ and only the second
 1641 line survives.

1642 *The interfaces.* $\mathbf{R}_A$, $\mathbf{R}_B$ and the transmission matrices into and out of the
 1643 layer follow from continuity of displacement and traction at each interface:
 1644 six equations in the states of the six modes on either side, solved for each
 1645 lateral wavenumber, evanescent orders included.

1646 *The incident field and the output.* A plane wave of unit amplitude arrives
 1647 from the upper half-space. Its field in the layer, a downgoing part and an
 1648 upgoing part with the reverberation summed as in Eq. (C.1), is the right-
 1649 hand side of the scheme. The output is the upgoing wave of the same type
 1650 as the incident one, arriving at z_A from the voxels and transmitted into the
 1651 upper half-space: the change that the layer of voxels makes to the same-type
 1652 reflection coefficient (R_{PP} , R_{SS} or R_{SH}), a quantity that does not depend on
 1653 how the waves are normalised.

1654 *The reference, and a check of the construction.* The exact change is the

1655 same-type reflection coefficient of the stack with the layer, less that of the
 1656 background, both from Kennett’s recursion [67], an independent algorithm.
 1657 Before any voxel is introduced, the background’s own reflection coefficient
 1658 computed from the modes and the interface matrices above agrees with Ken-
 1659 nett’s to 5×10^{-15} in all twelve cases.

1660 It is computed by two independent algorithms. The first is invariant
 1661 imbedding, the impedance march of Haines et al. [69]: the six-component
 1662 state of displacement and traction obeys $dy/dz = A(z)y$, and with the four
 1663 3×3 blocks A_{ij} of A the impedance Y , which gives the traction from the
 1664 displacement, obeys the Riccati equation

$$\frac{dY}{dz} = A_{21} + A_{22}Y - YA_{11} - YA_{12}Y. \quad (C.2)$$

1665 It is started below the stack with the impedance of the downgoing waves
 1666 of the lower half-space, which is the radiation condition, marched upwards,
 1667 exactly through the homogeneous gaps and by fourth-order Runge–Kutta
 1668 with the medium sampled at the stages through the graded layer, and at
 1669 the top the reflection matrix follows from continuity of displacement and
 1670 traction with the modes of the upper half-space. Y stays bounded, so no
 1671 growing exponential is formed. The second is the propagator: the same
 1672 state is integrated across the graded layer as a 6×6 matrix, with $A(z) =$
 1673 $F(z)\text{diag}(i\gamma_m)F(z)^{-1}$ built at each depth from the plane-wave modes of the
 1674 medium there, F the matrix of the six modes’ displacements and tractions
 1675 and γ_m their vertical wavenumbers, and the radiation conditions of the two
 1676 half-spaces are imposed on the result. For the graded layer the two agree
 1677 to 10^{-12} in all twelve cases with 400 steps of the march, whose error falls
 1678 sixteenfold for each doubling of the steps; for a uniform layer the propagator

1679 equals Kennett's recursion to 3×10^{-14} .

1680 *Oblique incidence, and a three-layer background.* At oblique incidence the
1681 test is the three-dimensional scheme of Eq. (B.1), with P, SV and SH waves,
1682 in two backgrounds: the two half-spaces above, and a three-layer background
1683 in which the voxels lie in a layer between a slower half-space above ($\alpha = 4200$,
1684 $\beta = 2400 \text{ m s}^{-1}$, $\rho = 2300 \text{ kg m}^{-3}$) and the faster one below (6500, 3700,
1685 2800), so that the layer reverberates between two interfaces. The construc-
1686 tion is the one set out above: the reverberation added to the whole-space
1687 kernel at every order of the reciprocal lattice, Eq. (C.1), the interface ma-
1688 trices, the incident field, and the reference, Kennett's recursion [67], against
1689 which the background's own reflection coefficient is reproduced to 5×10^{-15}
1690 in all twelve cases before any voxel is introduced.

1691 The results are in Table C.16, at $\omega = 3000 \text{ rad s}^{-1}$ ($k_S D = 2$; at 300 rad s^{-1}
1692 the sixth-order errors lie below round-off, as in Table B.15) and 20° in the
1693 upper half-space. In all thirty-six cases the order is that of the whole space,
1694 with the interfaces 3m from the layer and with the voxels against them
1695 (`scripts/measure_oblique_stratified_background.py`). The errors are
1696 close to those of the whole space at the same frequency (Table B.15): the
1697 background changes the field the voxels must represent, not how well they
1698 represent it.

1699 *The two terms of the error at oblique incidence.* The orders say that the
1700 scheme converges; the two-term analysis says what its error is. Both terms
1701 are tested at 20° , in the whole space and in the stratified backgrounds, for
1702 P, SV and SH waves.

Table C.16: The layer of voxels in a stratified background at 20° , $\omega = 3000 \text{ rad s}^{-1}$: relative error of the change of the same-type reflection coefficient on $n = 4$ planes, and the order over the last halving ($n = 2 \rightarrow 4$), for field degrees $p = 0, 1$ and 2 .

background	wave	$p = 0$	$p = 1$	$p = 2$
two half-spaces, interface 3 m below	P	6.1×10^{-3} (2.03)	2.7×10^{-6} (4.02)	2.3×10^{-9} (6.04)
	SV	1.5×10^{-2} (2.11)	2.6×10^{-5} (4.13)	3.7×10^{-8} (6.09)
	SH	1.6×10^{-2} (2.11)	2.7×10^{-5} (4.13)	4.1×10^{-8} (6.09)
two half-spaces, interface against the voxels	P	5.1×10^{-3} (2.04)	3.8×10^{-6} (4.04)	2.1×10^{-9} (6.00)
	SV	1.6×10^{-2} (2.10)	2.5×10^{-5} (4.14)	3.3×10^{-8} (6.11)
	SH	1.7×10^{-2} (2.10)	2.4×10^{-5} (4.15)	3.4×10^{-8} (6.11)
three layers, interfaces 3 m away	P	6.3×10^{-3} (2.03)	8.7×10^{-7} (4.04)	2.4×10^{-9} (6.06)
	SV	1.0×10^{-2} (2.10)	5.8×10^{-6} (4.08)	5.2×10^{-8} (6.11)
	SH	1.3×10^{-2} (2.10)	3.0×10^{-6} (3.95)	9.3×10^{-8} (6.10)
three layers, interfaces against the voxels	P	4.8×10^{-3} (2.04)	2.4×10^{-6} (4.04)	4.1×10^{-9} (6.03)
	SV	8.5×10^{-3} (2.09)	7.1×10^{-6} (4.01)	5.0×10^{-8} (6.09)
	SH	1.4×10^{-2} (2.10)	5.8×10^{-6} (3.92)	8.5×10^{-8} (6.12)

1703 *The first-order term.* The scheme is written as $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$ with the
 1704 output $u = \mathbf{r} \mathbf{c}$, as in Eq. (G.8), so that $T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}$ with no differencing.
 1705 The prediction is built from plane waves. The background field in the layer
 1706 is a sum of six, P, SV and SH going down and going up, and the voxels
 1707 radiate into six. For a pair made of an incident wave of wavevector $\mathbf{k}_{\text{in}}$ and
 1708 a radiated wave of wavevector $\mathbf{k}_{\text{out}}$, a cubic voxel of half-side h whose field is
 1709 held to total degree p gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (\text{C.3})$$

1710 the sum being over the Legendre products $P_{a_1} P_{a_2} P_{a_3}$ of the voxel's basis and
 1711 j_n the spherical Bessel functions: the numerator is the projection of the inci-
 1712 dent wave onto the basis paired with that of the outgoing wave, the denomi-
 1713 nator the exact pairing. Its leading departure from one is $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$ for
 1714 $p = 0$, with $d = 2h$, and along one axis it is the factor of Eq. (39). The pre-
 1715 dicted first-order term is the sum over the thirty-six pairs of the exact term
 1716 of each pair, an integral of two exponentials over the voxel, times its factor,
 1717 the radiated waves then being carried to the observer by the reverberation
 1718 of Eq. (C.1) exactly. On one and two planes of voxels of degree 0, 1 and
 1719 2, at $\omega = 3000 \text{ rad s}^{-1}$, the scheme's first-order term equals this prediction
 1720 to 2×10^{-15} in all twelve combinations of background and wave type. The
 1721 first-order error is therefore known in closed form at oblique incidence, in a
 1722 stratified background, for every wave type.

1723 *The second-order term.* For the law of Eq. (44) the contrast must vary

1724 within the voxels. Each plane of voxels is given the smooth profile of §9
 1725 in depth, projected to Legendre degree $r = p$, and the coupling is that of
 1726 Eq. (41) with the reverberation added. The second-order term of the scheme
 1727 is $T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}$. The exact T_2 needs the exact stack with a layer
 1728 whose properties vary continuously in depth, at oblique incidence, for which
 1729 a recursion over homogeneous layers is not exact. It is computed by two
 1730 independent algorithms, invariant imbedding [69] and the propagator of the
 1731 six-component state, which agree to 10^{-12} in all twelve cases (above). T_2 then
 1732 follows from Eq. (G.9) at $\epsilon = 0.1$, where halving ϵ changes it by 2×10^{-8} .
 1733 The ratio of the relative error of T_2 to the relative projection error $\mathcal{E}_p$ is in
 1734 Table C.17. On four planes the law holds to 3% for $p = 0$ and 1 and to 6%
 1735 for $p = 2$ at $k_S D = 0.2$, and to 0.6% for all three degrees at $k_S D = 0.067$, in
 1736 every background and for every wave type: the departure falls about ninefold
 1737 as the frequency falls threefold, as $(kD)^2$. On two planes the ratios for $p = 0$
 1738 and 1 lie within 0.03 of those in the table; for $p = 2$ they are lower, 0.89 to 0.99
 1739 at $k_S D = 0.2$ and 0.988 to 0.999 at $k_S D = 0.067$. The law is loosest for the
 1740 highest degree at the highest frequency, because the projection error of degree
 1741 2 is so small (1.6×10^{-5} on four planes) that the departure from the long-wave
 1742 limit is a larger share of it. Both terms of the error are thus known in advance
 1743 at oblique incidence and in a stratified background, as they are at normal
 1744 incidence in the whole space (`scripts/measure_oblique_two_term.py`).

1745 Two limits stand. The oblique construction is implemented once, and
 1746 its checks are the exact solutions. And the voxels here fill their plane, so
 1747 that the image of a voxel in an interface it touches is the image of a whole
 1748 plane; a body of finite lateral extent against an interface, where that image

Table C.17: The relative error of the second-order term over the relative projection error of the medium, at 20° , on $n = 4$ planes with $r = p$; the law of Eq. (44) predicts one in the long-wave limit.

		$k_S D = 0.2$			$k_S D = 0.067$		
background	wave	$p = 0$	$p = 1$	$p = 2$	$p = 0$	$p = 1$	$p = 2$
whole space	P	1.000	0.998	1.011	1.000	1.000	1.000
	SV	0.998	0.992	1.032	1.000	0.999	1.004
	SH	0.999	0.992	1.031	1.000	0.999	1.003
two half-spaces, interface 3 m below	P	1.010	1.009	1.019	1.001	1.001	1.001
	SV	1.012	1.008	1.046	1.001	1.001	1.005
	SH	1.027	1.025	1.059	1.003	1.003	1.006
three layers, interfaces 3 m away	P	1.004	1.003	1.013	1.000	1.000	1.000
	SV	0.979	0.973	1.002	0.997	0.996	1.000
	SH	1.007	1.004	1.035	1.001	1.001	1.004
three layers, interfaces against the voxels	P	1.002	1.001	1.011	1.000	1.000	1.000
	SV	0.995	0.988	1.018	0.999	0.999	1.002
	SH	1.004	1.002	1.032	1.000	1.000	1.003

1749 is singular, has not been run.

1750 **Appendix D. The sphere: near field, an impedance march and a** 1751 **sharp surface**

1752 *The near field.* A receiver is seldom at infinity, and the scheme does not
1753 require it to be. The scattered field at a point $\mathbf{x}$ outside the voxels is the
1754 propagator applied to the source that each voxel carries, the contrast times
1755 the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \boldsymbol{\Delta}(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (\text{D.1})$$

1756 with $\mathbf{K}$ the nine-component Green's tensor of §2, the same one that couples
1757 the voxels in the solve, $\boldsymbol{\Delta}$ the contrast operator of Eq. (5) and e_j the poly-
1758 nomial internal state of voxel j . Its first three rows are the displacement and
1759 its last six the strain. Nothing is dropped: the terms in $1/r^2$ and $1/r^3$ are in
1760 $\mathbf{K}$. The integrand's source is a polynomial in each voxel and is integrated by
1761 a tensor Gauss rule with six nodes a side; the rule's own error is measured
1762 by repeating the readout with four nodes, and from 8 voxels across the two
1763 differ by less than 4×10^{-6} at $1.25a$ and by less than 10^{-10} at $2a$. The ob-
1764 servers are the nine angles of Fig. 6(b) on spheres of radius $1.25a$, $2a$ and
1765 $10a$, where $k_S r$ runs from 0.6 to 10; the reference is the exact partial-wave
1766 displacement at the same points; and the error is the measure of the far-field
1767 test. Table D.18 gives it for the mean-only voxel and for the first-moment
1768 voxel (Appendix Appendix E). The orders of the far field hold in the near
1769 field: second for the mean-only voxel at every distance and both frequencies,
1770 and 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the

Table D.18: The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

n_{sub}	mean only			first moment		
	1.25a	2a	10a	1.25a	2a	10a
$k_S a = 0.5$						
4	3.3×10^{-2}	1.2×10^{-2}	5.3×10^{-3}	1.4×10^{-2}	1.6×10^{-3}	6.3×10^{-4}
6	1.4×10^{-2}	5.2×10^{-3}	1.9×10^{-3}	1.2×10^{-3}	3.8×10^{-4}	1.9×10^{-4}
8	8.0×10^{-3}	2.9×10^{-3}	1.1×10^{-3}	4.7×10^{-4}	1.4×10^{-4}	7.3×10^{-5}
12	3.5×10^{-3}	1.2×10^{-3}	4.7×10^{-4}	7.8×10^{-5}	3.1×10^{-5}	1.7×10^{-5}
16	2.0×10^{-3}	6.7×10^{-4}	2.6×10^{-4}	2.5×10^{-5}	1.0×10^{-5}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.03	2.06	2.02	3.95	3.85	3.77
$k_S a = 1.0$						
4	4.8×10^{-2}	3.5×10^{-2}	3.3×10^{-2}	9.9×10^{-3}	1.3×10^{-3}	6.0×10^{-4}
6	2.0×10^{-2}	1.4×10^{-2}	1.4×10^{-2}	9.3×10^{-4}	3.0×10^{-4}	2.0×10^{-4}
8	1.2×10^{-2}	8.1×10^{-3}	7.8×10^{-3}	3.5×10^{-4}	1.1×10^{-4}	7.4×10^{-5}
12	5.2×10^{-3}	3.6×10^{-3}	3.5×10^{-3}	6.0×10^{-5}	2.5×10^{-5}	1.7×10^{-5}
16	2.9×10^{-3}	2.0×10^{-3}	1.9×10^{-3}	1.9×10^{-5}	8.3×10^{-6}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.01	2.01	2.00	3.95	3.84	3.78

1771 far field at this contrast. A quarter of a radius from the surface, with 16 vox-
 1772 els across, the first-moment voxel's error is 2.5×10^{-5} and 1.9×10^{-5} , 78 and
 1773 153 times below the mean-only voxel's. The error grows towards the body,
 1774 by a factor of 1.5 to 7 between $10a$ and $1.25a$, because the near field carries
 1775 the higher multipoles of the source, which a given grid resolves less well; the
 1776 order does not change. On the coarsest grid a voxel is as wide as the graded
 1777 shell and twice as wide as the gap to the nearest observer, and the sequence
 1778 is not yet asymptotic. Read at these distances with the asymptotic formula,
 1779 the same solutions would be in error by 48% to 85%: the far-field readout
 1780 is for the far field (`scripts/measure_graded_sphere_near_field.py`; the
 1781 readout is `graded_field` of the package, tested against the propagator of a
 1782 point source, against the gradient of its own displacement, and against the
 1783 far-field readout at great distance).

1784 *The voxels against an impedance march.* A second route to the same body
 1785 is invariant imbedding. The impedance of §9, Eq. (C.2), is marched up-
 1786 wards through the depth range of the sphere with the lateral variation of the
 1787 medium carried inside the march, on a laterally periodic grid, and a second
 1788 sweep carries the field down [69]; the march returns the reflected plane waves
 1789 on a plane above the body and the transmitted ones on a plane below. It is
 1790 run here on the graded sphere and set beside the voxels, to ask how many
 1791 voxels a given accuracy costs against an established alternative. Four things
 1792 are fixed so that the comparison is of like with like.

- 1793 1. *The planes.* Both methods are read on the planes $z = -1.25a$ and
 1794 $z = +1.25a$. On the tangent planes $z = \pm a$ themselves the observers lie
 1795 on the faces of the voxels at the poles, where the Gauss rule of the voxel

Table D.19: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

1796 readout (Eq. (D.1)) cannot integrate the singular kernel: its change
1797 from four to six nodes is 2% to 10% there, more than the scheme's error,
1798 and below 10^{-4} a quarter of a radius away. The march's amplitudes are
1799 carried across the same homogeneous gap by each order's own phase,
1800 which is exact.

1801 2. *The periodic array.* The march needs a laterally periodic grid, so it
1802 solves a square array of spheres. Its answer is scored against the exact
1803 solution of one sphere summed over the array, by Poisson summation,
1804 which is exact but for scattering between the spheres. That part, and
1805 the lateral resolution, are errors of the method as used on a single body,
1806 and are counted.

1807 3. *The error.* For the march, the norm of the difference over the plane-

1808 wave orders of the grid, relative to the norm of the exact scattered
1809 orders; for the voxels, the norm of the difference of the scattered dis-
1810 placement over a patch of side $5a$ on each plane, relative to the norm
1811 of the exact one. The two are the same quantity in wavenumber and
1812 in space, for the array and for the single body.

1813 4. *The refinement.* For the march, the lateral grid, with the depth step
1814 refined until the error no longer changes; for the voxels, the grid and
1815 the degree.

Table D.20: The graded sphere at $k_S a = 0.5$ read on the planes $z = \mp 1.25a$: relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is $5a$ unless stated.

method	setting	unknowns	reflection	transmission
impedance march	12×12 , 16 steps	432	1.2×10^{-2}	9.9×10^{-3}
	16×16 , 32 steps	768	4.3×10^{-3}	3.6×10^{-3}
	20×20 , 32 steps	1200	7.6×10^{-4}	7.3×10^{-4}
	28×28 , 48 steps	2352	2.5×10^{-4}	5.6×10^{-4}
	30×30 , period $7.5a$	2700	7.2×10^{-4}	4.7×10^{-4}
mean-only voxel	8 across	3672	5.0×10^{-3}	7.9×10^{-3}
	16 across	24 984	1.2×10^{-3}	2.0×10^{-3}
first-moment voxel	6 across	7488	3.3×10^{-4}	5.2×10^{-4}
	8 across	14 688	1.4×10^{-4}	2.2×10^{-4}
	16 across	99 936	1.1×10^{-5}	1.8×10^{-5}
second-moment voxel	8 across	36 720	1.4×10^{-5}	2.3×10^{-5}
	10 across	69 120	2.6×10^{-6}	4.2×10^{-6}

1816 Table D.20 gives the result. The march converges quickly in the lateral grid,

1817 because the body is smooth, and with far fewer unknowns than any voxel
 1818 scheme: at 20×20 it is at 7×10^{-4} with 1200 unknowns on a plane. The
 1819 mean-only voxel does not reach that with sixteen voxels across and twenty
 1820 times the unknowns; on a sharp sphere, where it also carries a staircase, it
 1821 fares worse. The first-moment voxel reaches the march's accuracy with six
 1822 voxels across, and below about 2×10^{-4} it continues at close to fourth order
 1823 where the march does not: in transmission the march's error at a period
 1824 of $5a$ stops near 5×10^{-4} , which is the scattering between the spheres of
 1825 its array, since lengthening the period lowers it and refining the grid does
 1826 not. The costs are of different kinds. The march's unknowns are few but its
 1827 operator is a dense matrix on the plane, marched through the depth, and a
 1828 finer lateral grid requires a smaller depth step (with eight steps the 20×20
 1829 march diverged); the voxel schemes have many unknowns and a product done
 1830 by fast Fourier transform. In the timings of these runs, taken on one machine
 1831 with other work in progress and therefore only indicative, the first-moment
 1832 voxel at six across was solved in about a sixth of the time of the 20×20 march.
 1833 The march has what the comparison does not use: it returns the reflection
 1834 and transmission for every incident plane wave of the grid at once, and a
 1835 laterally periodic or laterally extended medium is its natural problem and
 1836 not a cost. The comparison is at one frequency (`scripts/measure_graded_`
 1837 `sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

1838 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 1839 and its voxelisation changes with every grid. Table D.21 shows its error for
 1840 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_S a = 0.5$.
 1841 The error tracks the staircase's volume, which is 5%, 17% and 4% too large

1842 on these grids. Rescaled by the ratio of the sphere’s volume to the staircase’s,
1843 it falls below 1% at $k_S a = 0.5$ and to at most 3% at $k_S a = 1.0$, but it still
1844 does not converge: what remains is the staircase’s shape. A refinement study
1845 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
1846 measures the scheme.

Table D.21: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_S a = 0.5$		$k_S a = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

1847 Appendix E. The three-dimensional voxel solver

1848 The three-dimensional results of §§10 and 11 use the Legendre voxels
1849 of §8 on a cubic grid, with the following construction (`cubic_scattering/`
1850 `graded_voxel`).

1851 *The voxel.* Each cell carries the nine-component state of §2, displacement
1852 and engineering strain, with the strain an *independent* unknown and not the
1853 gradient of the displacement: a strain derived from a cell’s projected dis-
1854 placement jumps across the faces between cells and leaves a spurious source
1855 there (§8). The state is expanded in the Legendre products of degree at

most p in the cell's local coordinates $\boldsymbol{\xi} \in [-1, 1]^3$, and the contrast operator $\boldsymbol{\Delta}$ in those of degree at most r , its coefficients the L^2 projection of the medium; $p = r = 1$ gives the first-moment voxel of 36 weights. The source $\boldsymbol{\Delta}\psi$ is then a polynomial of degree $p + r$, expanded in monomials m_c . The Lippmann–Schwinger equation is tested with each basis function over each cell (Galerkin), and cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\boldsymbol{\xi}) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\boldsymbol{\xi}') d^3x d^3x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (\text{E.1})$$

with $\mathbf{P}$ the 9×9 point propagator of §2.

One integral in place of two. With $\mathbf{s} = \mathbf{u} - \mathbf{u}'$ the difference of the two local positions, Eq. (E.1) becomes $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s$ over $\mathbf{s} \in [-2h, 2h]^3$, where W_{ac} is a product over the three axes of one-dimensional cross-correlations of the test and source polynomials: continuous, polynomial on $[-2h, 0]$ and $[0, 2h]$, with kinks at 0 and $\pm 2h$. For cells that do not touch, the kernel is smooth on each of the eight pieces, and a three-dimensional Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional quadrature of Eq. (E.1) to 10^{-9} one cell beyond contact.

The self cell and its 26 neighbours. The static part of $\mathbf{P}$ is a sum of derivatives of r^{-1} and r , distributions at the origin. Derivatives are moved onto W until the kernel is at most as singular as $1/r$, and the kinks of W then carry plane δ functions; every piece is a box or a rectangle with its corners on the breakpoints, so the singular point is either a vertex of a piece, integrated with Duffy's pyramids, or at least $2h$ away. The δ function of the inclusion problem is carried by those plane δ functions and is not added by hand.

1879 The construction reproduces the sum rule $\sum_p \int_V \int_V \partial_p \partial_p |\mathbf{x} - \mathbf{x}'|^{-1} d^3x d^3x' =$
1880 $-4\pi V$, the biharmonic rule for r and the cube's Coulomb self-energy to 10^{-11} ,
1881 and every block equals the sum of the 64 blocks between the two cells' half-
1882 size sub-cells, with the polynomials re-expressed on them, to 2×10^{-15} . The
1883 dynamic remainder of $\mathbf{P}$ is at most weakly singular and is integrated by the
1884 Gauss rule of the pieces.

1885 *Symmetry, and the solve.* A block depends only on the offset $\mathbf{o}$ between the
1886 two cells. The background is isotropic and the cube is mapped to itself
1887 by its 48 signed permutations $\mathbf{Q}$, under which $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^\top$, with
1888 $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$ and $\mathbf{R}_6$ the representation of $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^\top$ on Voigt
1889 vectors. With the test and source polynomials transforming by matrices $\mathbf{T}$
1890 and $\mathbf{U}$, $\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^\top$, so one block is computed for each
1891 orbit of offsets and the rest follow exactly. The coupling is then a convolution
1892 over the grid, applied by the fast Fourier transform on a $(2n-1)^3$ circular
1893 embedding, and the system is solved by GMRES preconditioned with each
1894 cell's own block. Where both can be run, the solution equals that of a dense
1895 solve to 10^{-10} relative (`tests/test_graded_voxel_fft.py`). Every cell that
1896 overlaps the body is kept, each with the L^2 projection of its contrast by a
1897 six-point Gauss rule per axis; keeping only the cells whose centre lies inside
1898 the body drops the contrast of the partly filled cells, and the mean-only
1899 voxel then converges at order one instead of two. The far field is read at
1900 5×10^8 radii, where the near-field terms that the asymptotic form omits are
1901 negligible.

1902 *The first-moment voxel on the graded sphere.* The voxel with $p = r = 1$ is
1903 applied to the graded sphere of §10. To isolate the discretisation error from

1904 the rescattering the grid cannot yet resolve, the contrast is first scaled by
 1905 10^{-6} , and the exact sphere is solved at the same scaled contrast, so that no
 1906 Born approximation enters the reference: the relative error is then that of
 1907 the scheme's first-order term, in which the field is the incident one carried
 1908 across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the
 1909 error is 7.5×10^{-5} , 5.0×10^{-6} and 1.6×10^{-6} at $k_S a = 0.5$, and 2.3×10^{-4} ,
 1910 9.1×10^{-5} and 2.8×10^{-5} at $k_S a = 1$. The apparent orders between 6 and
 1911 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a
 1912 voxel at $n = 4$ being as wide as the graded shell (`scripts/pilot_graded_`
 1913 `voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16
 1914 voxels across, where the error is 4.9×10^{-6} at $k_S a = 0.5$ and 5.9×10^{-6} at
 1915 $k_S a = 1$, and the apparent order approaches four at a rate set by the number
 1916 of voxels across the graded shell.

1917 **Appendix F. The uniform-strain closure against the exact sphere**

1918 *Against the exact sphere.* These statements are tested against the exact elas-
 1919 tic Mie solution for a sphere, with no Eshelby formula in between. The
 1920 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 1921 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 1922 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 1923 route of Eq. (26) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 1924 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 1925 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 1926 isotropic shear average, Eq. (32), is the depolarisation read from Mie's E_2
 1927 to 10^{-8} . The same moment taken as a principal value, over the ball with

1928 a small sphere excluded around the origin, has zero depolarisation in every
 1929 channel. The whole static response of the sphere is the δ term.

1930 At finite frequency the closure uses the dynamic ball moments, which are
 1931 in closed form. It departs from Mie at second order in ka and first order in
 1932 the contrast, and the departure is closed-form too: the ball's moments are
 1933 elementary, and Mie's low-frequency series with the contrast kept exact gives
 1934 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (\text{F.1})$$

1935 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 1936 the radiation damping, agree exactly in all three channels. So the closure's
 1937 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 1938 own. That is the order the hierarchy assigns to the omitted second gradi-
 1939 ents (§6.3), and it has the same origin, the variation of the field across the
 1940 scatterer, as the collocation error of §7. Eq. (F.1) is derived symbolically
 1941 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 1942 the numerical Mie solve and an independent implementation of the closure
 1943 to 5×10^{-4} , which is the size of the next order in the contrast.

1944 **Appendix G. The scattering series of the layer, and the projection-** 1945 **error law**

1946 *The scattering series.* Multiply the whole contrast, $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$
 1947 together, by a number s , and let $u(s)$ be the scattered displacement at an

1948 observer at depth z_o outside the layer. It vanishes at $s = 0$ and is analytic
 1949 in s there, and its expansion is the Born series,

$$u(s) = s T_1 + s^2 T_2 + s^3 T_3 + \dots, \quad (\text{G.1})$$

1950 so that T_n is the part of the field at the full contrast, $s = 1$, that has been
 1951 scattered n times.

1952 *The first two terms of the exact layer.* They are integrals over the layer.
 1953 With the incident state $w^0 = (u^0, \varepsilon^0)$, the contrast $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$
 1954 with profile $f(z)$, and the kernel $\mathbf{K}$ of Eq. (38), whose first row $(g, g')(z_o - z)$
 1955 gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (\text{G.2})$$

1956

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (\text{G.3})$$

1957 Here w^1 is the state scattered once, inside the layer; its strain includes the
 1958 local term of the kernel, $-f(z) \Delta M \varepsilon^0(z)/M_P$. Each further term repeats
 1959 the step, $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$. For a plane force at z_s
 1960 above the layer, $w^0(z) = c e^{ik(z-z_s)}(1, ik)$ with $c = i/(2M_P k)$, every integrand
 1961 is an exponential times the profile. Writing $a = \omega^2 \Delta \rho$ and $b = \Delta M$, and
 1962 carrying out the products of Eqs. (G.2) and (G.3), the terms seen in reflection
 1963 ($z_o < 0$) and in transmission ($z_o > D$) are, for any profile,

$$T_1^R = c^2 e^{-ik(z_o + z_s)} (a + bk^2) \int_0^D f(z) e^{2ikz} dz, \quad T_1^T = c^2 e^{ik(z_o - z_s)} (a - bk^2) \int_0^D f(z) dz, \quad (\text{G.4})$$

1964

$$\begin{aligned}
T_2^R &= c^2 e^{-ik(z_o+z_s)} \left[c(a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 e^{2ikz} dz \right], \\
T_2^T &= c^2 e^{ik(z_o-z_s)} \left[c(a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c(a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\
&\quad \left. + \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 dz \right].
\end{aligned} \tag{G.5}$$

1965 The first-order reflection is the Fourier transform of the profile at the wavenum-
1966 ber $2k$, and the first-order transmission its integral. In each second-order
1967 term the last integral is the local term of the kernel, in which both scat-
1968 terings are at one point and the profile appears squared; the others are the
1969 two scatterings at different depths. In transmission the wave scattered twice
1970 forwards carries the contrast of velocity twice, $(a - bk^2)^2$, and the wave scat-
1971 tered backwards and then forwards again the contrast of impedance twice,
1972 $(a + bk^2)^2$. For the uniform layer ($f = 1$) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a + bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a - bk^2) D, \tag{G.6}$$

1973

$$\begin{aligned}
T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_P} \left[-i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\
T_2^T &= \frac{i c^2 e^{ik(z_o-z_s)}}{8k^3 M_P} \left[a^2(1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2(e^{2ikD} - 1 + 2kD(kD - i)) \right. \\
&\quad \left. + b^2 k^4(1 - e^{2ikD} + 2kD(kD - 3i)) \right].
\end{aligned} \tag{G.7}$$

1974 The reflection sees the contrast of impedance, $a + bk^2 = k^2 M_P(\Delta\rho/\rho +$
1975 $\Delta M/M_P)$, and the transmission that of velocity, $a - bk^2$. As $kD \rightarrow 0$, $T_2^R \rightarrow$
1976 $-c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_P$: the local term alone, which is the starting point of
1977 the argument at the end of this appendix. For the smooth profile the integrals

1978 of Eqs. (G.4) and (G.5) are evaluated in closed form symbolically, and by
 1979 quadrature they return the same values to 10^{-16} . Equations (G.6) and (G.7)
 1980 equal, identically, the first two Taylor coefficients in s of the exact reflection
 1981 and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At
 1982 $\omega = 300 \text{ rad s}^{-1}$, $|T_2|/|T_1|$ is 0.039 in reflection and 0.17 in transmission for
 1983 the uniform layer.

1984 *The first two terms of the scheme.* Write Eq. (38) as $(\mathbf{M} - \mathbf{K}) \mathbf{c} = \mathbf{b}$, with
 1985 $\mathbf{M}$ the diagonal matrix of the $\int_i \phi_a^2 dz$, $\mathbf{K}$ the matrix of the coupling blocks
 1986 times Δ_1 , and $\mathbf{b}$ the moments of the incident field, and let $\mathbf{r}$ be the row that
 1987 gives the scattered displacement at the observer from the coefficients, $u = \mathbf{r} \mathbf{c}$.
 1988 Scaling the contrast by s scales $\mathbf{K}$ and $\mathbf{r}$ by s , so $u(s) = s \mathbf{r} (\mathbf{M} - s \mathbf{K})^{-1} \mathbf{b}$ and

$$T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}, \quad T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}. \quad (\text{G.8})$$

1989 $\mathbf{M}^{-1} \mathbf{b}$ is the projection of the incident field onto each cell's functions: the first
 1990 term needs no solve. Splitting $\mathbf{K}$ into its blocks within a cell and its blocks
 1991 between cells splits T_2^{scheme} into the part scattered twice in one cell and the
 1992 part scattered in two; the same split of the inner integral of Eq. (G.3) splits
 1993 the exact T_2 .

1994 *A check by differences.* The terms can also be read from solutions at four
 1995 scaled contrasts, $s = \pm\epsilon$ and $\pm 2\epsilon$, with no knowledge of the formulas above.
 1996 The odd and even parts of u in s separate the odd orders from the even,
 1997 and the two step sizes remove the next term of each (the five-point central
 1998 differences for $u'(0)$ and $\frac{1}{2}u''(0)$):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (\text{G.9})$$

1999 A negative s is a medium slower and lighter than the background by the
2000 same fractions, as legitimate a medium as the other. With $\epsilon = 0.01$ the
2001 remainders $4\epsilon^4 T_5/T_1$ and $4\epsilon^4 T_6/T_2$ are below 2×10^{-12} for this layer, and the
2002 formulas applied to the exact layer return Eqs. (G.6) and (G.7) less exactly
2003 those remainders, to 10^{-17} . The measurements of §9 use the differences for
2004 the scheme and the exact layer alike, each in its own arithmetic, and they
2005 agree with Eqs. (G.2), (G.3) and (G.8).

2006 *Why the terms must be separated.* The law of Eq. (44) is a statement
2007 about T_2 . At the full contrast the terms of third and higher order are 7% of
2008 the second and carry errors of their own, and if every order above the first
2009 is taken together the ratio of Eq. (44) reads 0.90 to 0.94 where the second-
2010 order term alone gives 1.00 to 1.01. The shortfall is itself the single site's. In
2011 the long-wave limit the kernel is local, the exact strain is $\varepsilon_0/(1 - cf)$ with
2012 $c = -\Delta M/M_P$, and the scheme's is the Galerkin solution of $\varepsilon = \varepsilon_0 + cf\varepsilon$ in
2013 each cell on that cell's polynomials; the relative error of the term of order
2014 n is $(\int f^n dz - \int (\Pi f)^n dz)/\int f^n dz$, 2.455 times $\mathcal{E}_q$ for the third term of
2015 this profile, which has the opposite sign to the second. At $\omega = 30 \text{ rad s}^{-1}$
2016 the whole nonlinear part has 0.896 times $\mathcal{E}_q$, and the cell-by-cell Galerkin
2017 solution reproduces it to four digits with c from the contrast (`scripts/`
2018 `measure_layer_t3_low.py`).

2019 Equation (44) is exact in the long-wave limit, for a reason that shows what
2020 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
2021 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
2022 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
2023 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-

2024 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order
 2025 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f \, dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) \, dz = \|\Pi_q f\|^2, \quad (\text{G.10})$$

2026 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 2027 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 2028 scattering takes place in the cell of the first: T_2 is the second-order term of
 2029 the single-site T -matrix, and Eq. (44) is a statement about the single site.
 2030 The T -matrix sums the rescattering within a cell exactly for the cell as it
 2031 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 2032 cell is not the medium. The split is measured directly (`scripts/measure_`
 2033 `layer_bases.py`): at $\omega = 300 \text{ rad s}^{-1}$ on 4, 8 and 16 cells, T_2 of the scheme
 2034 and of the exact layer are each divided into the part whose two scatterings
 2035 lie in one cell and the part between different cells. For all nine pairs the
 2036 error of the same-cell part is 0.998 to 1.013 times $|T_2| \mathcal{E}_{\min(p, r)}$, and that of
 2037 the part between cells at most 0.002 times it. The coupling between cells
 2038 contributes only the departure from the limit. Measured, the ratio of the
 2039 two sides of Eq. (44) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} ,
 2040 4.3×10^{-3} , 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 :
 2041 as $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (G.10)
 2042 is verified symbolically for all nine pairs, and the projection errors are com-
 2043 puted exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each
 2044 halving of the cells.

Appendix H. Relation to other methods

Relation to the coupled-dipole method. The same partition appears in the coupled-dipole method as the choice of polarisability. The Clausius–Mossotti relation, the lattice dispersion relation [11], the polarisability taken from the exact static solution [12] and the integrated Green tensor [13] all adjust the single site so that the lattice reproduces the continuum. For space-filling cubes the adjustment is neither needed nor a free parameter: the tiling identity fixes the local field exactly, the self term of the collocation closure is the one that makes it cancel, and what is left is a closed-form error rather than an empirical one. That the method is a collocation of the volume integral equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (35) is the same statement for the elastic lattice with the self term evaluated over the finite cube, and it is what makes the error a midpoint-rule error. The filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-limit the source to the grid; the source-cell average does the complementary thing exactly, its form factor vanishing on the reciprocal lattice, and the spectral construction of Eq. (B.1) is the periodic coupled-dipole method of Chaumet et al. [16] with the Galerkin double average in place of the point dipole. The second-order convergence reported for the DDA by Yurkin et al. [3] corresponds to the uniform-field voxel’s, and the opposite-signed constants of Eq. (36) explain why a polarisability correction can improve one observable only at the expense of another, as the package’s own form-factor corrections do in reflection and transmission. Going beyond second order requires a richer internal field, which is the point of higher-order bases for volume integral equations [72, 2]; the first-moment voxel is the lowest such

2070 basis, and in its spectral form the Galerkin double average makes the lattice
 2071 sum converge without Ewald splitting.

2072 *Relation to the spectral element method.* The voxel of degree p is close kin to
 2073 the spectral element [73], which was taken up for seismic wave propagation in
 2074 the 1990s, is set out for three-dimensional models by Komatitsch and Tromp
 2075 [74], and is now a standard tool for it. Both divide the medium into hexahe-
 2076 dra, represent the field in each by polynomials of high degree, and determine
 2077 the coefficients by a Galerkin condition on a weak form. The spectral element
 2078 uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of
 2079 the element; the voxel uses the Legendre polynomials themselves. Four of
 2080 the present results have their counterparts there.

2081 1. *A diagonal mass matrix.* The spectral element obtains one by integrat-
 2082 ing at the Gauss–Lobatto–Legendre points, at which each interpolant
 2083 vanishes but one [74]. That rule is not exact for the product of two
 2084 interpolants, so that the diagonal form is reached, in their words, by a
 2085 process of subintegration, as in the finite-element methods that lump
 2086 the mass. Here the Gram matrix $\mathbf{M}$ of Eq. (G.8) is diagonal with no
 2087 approximation, because the Legendre functions are orthogonal on the
 2088 cell; this is why the first-order term needs no solve. The two routes
 2089 serve different ends: theirs an explicit march in time, ours the terms of
 2090 the scattering series.

2091 2. *The doubled order.* A scattered amplitude is a functional of the field,
 2092 and a Galerkin method gives a functional to twice the order of the field
 2093 itself. The order $2p + 2$ is this doubling: the leading error of Eq. (39) is
 2094 the product of the part of the incident wave that the cell’s polynomials

miss and the part of the outgoing wave that they miss, each of order h^{p+1} .

3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [74]. The rule $\min(2p+2, 2r+2)$ says why that is the right choice, and what a coarser representation of the medium would cost.

4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [74]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [74]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a cou-

2120 pling between every pair of voxels, and a background restricted to one whose
 2121 Green's tensor is known. It differs also from the boundary integral methods,
 2122 which are restricted to a finite number of homogeneous regions [74]: a vol-
 2123 ume integral equation admits a medium that varies within the scatterer. The
 2124 voxel scheme is therefore a candidate where the heterogeneity is localised in
 2125 a layered medium, and not a replacement for the spectral element in a model
 2126 that is heterogeneous throughout. A spectral-element mesh is designed by a
 2127 rule of resolution, about five grid points to the shortest wavelength at poly-
 2128 nomial degrees between five and ten [74]. The error analysis of this paper is
 2129 the counterpart of that rule for the integral formulation: it gives the size and
 2130 the degree of the voxels for a required accuracy in closed form, and it adds a
 2131 term, the projection error of the medium.

2132 **Appendix I. The cost of higher order, and what the error analysis** 2133 **offers a fast solver**

2134 *The cost of fourth order.* A first-moment voxel carries four times as many
 2135 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 2136 forms show what that buys. To reach an error ε in reflection, the uniform
 2137 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 2138 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 2139 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 2140 number of unknowns by about 5×10^4 (these are the Born-order constants;
 2141 at the contrast studied the first-moment voxel's measured errors lie within
 2142 5% of them, while the uniform voxel's transmission constant grows with the
 2143 contrast, so the comparison is if anything conservative). The stratified model

2144 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 2145 moment voxels for the same accuracy. Higher moments continue the trend
 2146 in one dimension: by Eq. (40) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 2147 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 2148 a quarter of a wavelength. In three dimensions, with the Legendre products
 2149 of total degree at most p , a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per
 2150 state component, against 4 for the first-moment voxel. The voxel of degree
 2151 2 converges at sixth order in the layer at oblique incidence (Table B.15);
 2152 that of degree 3 has not been built in three dimensions. The gain is largest
 2153 precisely where it matters: when the model varies on scales that are coarse
 2154 compared with the accuracy the wavefield must have.

2155 *What the error analysis offers a fast solver.* A voxel scheme is dense, and
 2156 for a large medium it is solved iteratively with a fast product. The adaptive
 2157 integral method [75] and the precorrected fast Fourier transform [76] are
 2158 the standard forms: the sources are projected onto a regular grid, in the
 2159 first by matching their multipole moments and in the second by matching
 2160 the potentials they produce at a set of test points, the interactions between
 2161 distant cells are evaluated on that grid by fast Fourier transforms, and those
 2162 between near cells are computed directly and corrected. Such a method has
 2163 three accuracies to set, that of the discretisation, that of the projection, and
 2164 the radius of the near zone, and they are usually set by trial. The results of
 2165 this paper set them beforehand, in five ways.

- 2166 1. *The target is known.* The discretisation error is known before the solve,
 2167 in order and in size: the first-order factor of a cell, Eq. (39) and its
 2168 form for any pair of wavenumbers (§9), and the projection error of the

medium, Eq. (44). A fast product need be no more accurate than that,
and it is known when it is accurate enough.

2. *The two errors do not mix.* The error of the scheme is the single site's:
the coupling between cells, averaged over the cells, is exact (§5). An
approximation of the far coupling to a set tolerance therefore adds its
own error to the discretisation's and does not alter it, so the two can
be budgeted separately.
3. *For cubes on a lattice there is nothing to project.* The cells are the reg-
ular grid. The coupling depends on the separation of the cells alone, so
the product is a convolution, block by block in the polynomial degrees,
and the fast Fourier transform applies to it exactly, as in Georgakis
et al. [2] for the linear voxel. What the projection onto the grid ap-
proximates in general is here an identity.
4. *For cells of several sizes the moments are already in hand.* Where the
cells differ in size or degree and are mapped to a uniform auxiliary
grid, the quantities to be matched are the polynomial moments of a
cell, which are the unknowns of the scheme; and the first-order factor
says how many of them matter at a given kd .
5. *A stratified background splits as the method requires.* The Green's ten-
sor of a plane-stratified background is the whole-space tensor, which is
invariant under translation in three dimensions and carries all the sin-
gularity, plus a reverberation, Eq. (C.1), which is smooth, is invariant
under lateral translation, and separates in depth into one-dimensional
moments. The first is the convolution above; the second is a fast Fourier
transform laterally and a product in depth, with no near-zone correc-

tion except where a voxel touches an interface. A splitting of the kernel into a term that carries the singularity and a nonsingular remainder is used to a related end by Sharma and Triverio [77], for surface integral equations in homogeneous and in layered backgrounds: there the singular term is independent of frequency, so that the direct near-zone integrals are computed once for a whole sweep in frequency. And because the orders and the error law are unchanged by the background (§9), the error budget of the whole space carries over to it.

No such solver is built here. These are consequences of the analysis, and they are to be tested.

Appendix J. The evidence for each result, and where to reproduce it

Table J.22 lists each result with the evidence for it. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, GradedContrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the moment integrals are `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python half of the last in `scripts/gate_cube_shear_split.py`; the stratified reference against Kennett’s recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_`

2218 `vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded,
 2219 with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded
 2220 sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with
 2221 the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_
 2222 sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_
 2223 vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.
 2224 wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the
 2225 independent implementation of the first-moment voxel is `scripts/crosscheck_
 2226 first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_
 2227 second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_
 2228 graded_contrast.py`; the projection-error law and its split are measured by
 2229 `scripts/measure_layer_bases.py`, in a stratified background by `scripts/
 2230 measure_layer_stratified_background.py`, `scripts/measure_oblique_
 2231 stratified_background.py` and `scripts/measure_oblique_two_term.py`,
 2232 and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_
 2233 degree.py`; the convergence figure is drawn by `scripts/plot_convergence_
 2234 orders.py`; the near field of the sphere is measured by `scripts/measure_
 2235 graded_sphere_near_field.py`; the moments of every grade are in `Mathematica/
 2236 CubeMomentHigherGrades.wl`, `Mathematica/CubeMomentStore.wl` and `Mathematica/
 2237 CubeMomentCanonical.wl`, with their independent evaluation in `scripts/
 2238 crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is
 2239 `Mathematica/ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_
 2240 layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_
 2241 ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.
 2242 py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All

²²⁴³ are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

Table J.22: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (26)	sum rule (27) exact; M, Q equal an independent archive, 81/81
Channels (29)–(31)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (32)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (F.1)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (35)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (36)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (38)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (B.1)	order 4, P and S, 20° and 40°

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2245 This work began thirty years ago in the author’s doctoral research at
2246 the Research School of Earth Sciences of the Australian National University,
2247 Canberra [21].

2248 **Declaration of competing interest**

2249 The author declares no known competing financial interests or personal
2250 relationships that could have appeared to influence the work reported in this
2251 paper.

2252 **Declaration of generative AI and AI-assisted technologies in the** 2253 **manuscript preparation process**

2254 *Software development..* Software development followed a human-directed,
2255 AI-assisted workflow. The author designed the software originally, but only
2256 directed and validated recent improvements, which build on the author’s
2257 original Fortran implementation (the `PhD_fortran_code` folder of the repos-
2258 itory given under Data availability) and an extensive library of Mathematica
2259 notebooks, developed and extended since 1992, that carry out and verify the
2260 symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus
2261 5.5) operated under human-in-the-loop supervision to implement specified
2262 changes and diagnostic tests. The design, the choice of numerical methods,
2263 and the resolution of the principal technical difficulties were the author’s. All
2264 AI-generated changes were subject to code review by the author and validated
2265 by automated tests and independent physical checks such as reciprocity, op-
2266 tical theorem, Hamiltonian symmetry, analytic limits and convergence.

2267 *Manuscript preparation..* During the preparation of this work, the author
2268 used Claude (Anthropic) in order to search for relevant open-access research,
2269 to write the TikZ code of the diagrams and to help redraft the paper into more
2270 coherent sections for readability. After using this tool, the author reviewed
2271 and edited the content as needed and takes full responsibility for the content
2272 of the published article.

2273 **Data availability**

2274 The Mathematica notebooks and Python scripts that derive and check
2275 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
2276 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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